2 Basic Laws, Nodal & Mesh Analysis

INTRODUCTION

In chapter 1, we have studied about basics of circuit elements. We also studied many circuit terminologies like current, voltage, power and energy etc. Now we are going to discuss some important laws which will help us for simplification of a circuit.

OHM'S LAW AND RESISTANCE

Ohm's law:

The current density in a conductor is proportional to the electric field intensity at a constant temperature. This is known as the point form of Ohm's law.

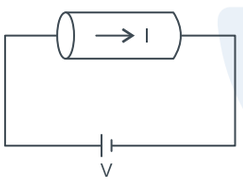


Fig. 2.1

$\vec{J} \propto \vec{E}$ (Temperature is constant)

$\vec{J} = \sigma \vec{E}$

Or "At constant temperature, the current through a conductor is proportional to the potential difference across its two ends".

$I \propto V$ [Temperature is constant]

$I = \frac{V}{R}$ [$\frac{1}{R}$ is a proportionality constant]

$R = \frac{V}{I}$ Volts/Ampere

Unit $\rightarrow$ Ohm (Ω)

R $\rightarrow$ Resistance

$G = \frac{1}{R}$ = Conductance

Definitions

Ohm's law: It states that the potential difference (V) between any two points on a passive linear circuit element are directly proportional to the current (I) flowing between these two points when temperature and other physical conditions are constant.

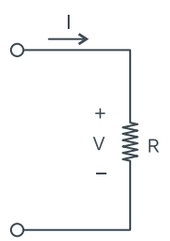


Fig. 2.2 Circuit Symbol for a Resistance

- **Ohm's law can also be written as:**
  I) $I = GV$
  II) $V = IR$
  These forms are applicable when Ohm's law is applied across a material.
- Current density $\propto$ electric field

$$J \propto E$$

$$J = \sigma E \rightarrow \text{Point form of Ohm's law}$$

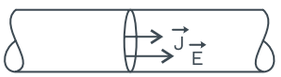


Fig. 2.3 Conductor Cross Section

- $\vec{J} = \sigma \vec{E}$

  This law can be applied at any point inside the material.
- σ = Proportionality constant = conductivity ($\Omega^{-1}\ m^{-1}$)
- ρ = resistivity (Ω – m)

$$\rho = \frac{1}{\sigma}$$

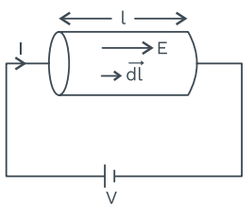


Fig. 2.4 A Voltage Applied Across Conductor

$$V = -\int \vec{E} \cdot \vec{dl}$$

$$V = E \times \text{length}$$

$$E = \frac{V}{l}$$

Current density = current per unit area (A/m^2)

$$J = \frac{I}{A}$$

Ohm's law: $J = \sigma E$

$$\frac{I}{A} = \sigma \times \frac{V}{l}$$

$$I = \frac{\sigma A}{l} V$$

Grey Matter Alert!

Electrical resistivity is the fundamental property of a material that measures how strongly it resists the electric current. It's inverse is called electrical conductivity which quantifies how well a material conducts electricity.

$$V = I\left(\frac{l}{\sigma A}\right) = I\left(\frac{\rho l}{A}\right) = IR$$

$V = IR$

Resistance, $R = \frac{\rho l}{A}$

where ρ = Resistivity of material (in Ω–m)
A = Area of cross-section (area perpendicular to current flow)
l = Length of material (Along the direction of current flow)

Note:

- Ohm's law states that the graph between voltage and current should be linear.

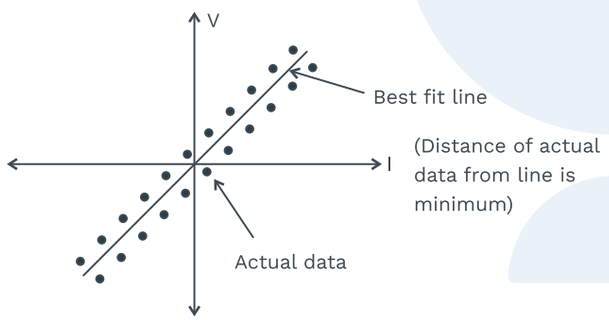


Fig. 2.5 Linear Graph

Definitions

Resistance: Resistance is the properties of a material due to which it opposes the flow of current through it.

Resistance (R):

- Resistance opposes the flow of current.
- Unit of resistance is ohms (Ω).
- Resistance of conductor materials is very small that which means opposition to the current flow is less.
- Mathematically, it is written as-

  Resistance, $R = \frac{\rho l}{A}$

  Where
  ρ = Resistivity of material in (Ω–m)
  A = Area of cross-section (Area perpendicular to current flow)
  l = Length of material (along the direction of current flow)

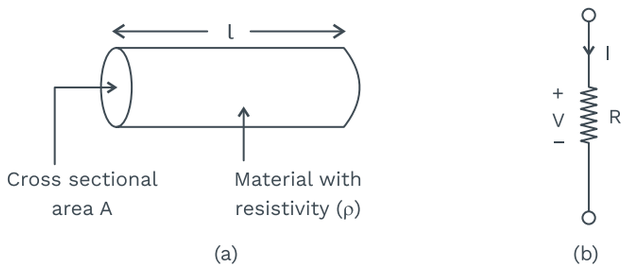


Fig. 2.6 Cylindrical Wire with Resistance R

Note:

- If resistance is independent of voltage and current, then it is called linear resistor.
- If resistance depends on voltage or current, then it is called non-linear resistor.

$R \propto l$ ... (i)

$R \propto \frac{1}{A}$ ... (ii)

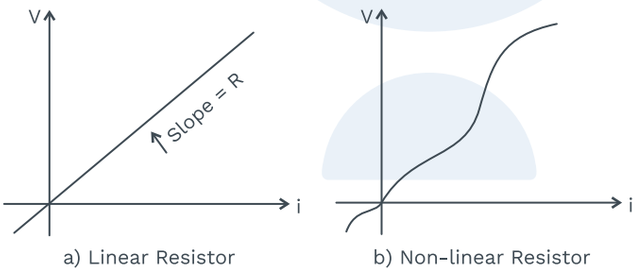


Fig. 2.7 i-V Characteristics of Resistors

Combining i) and ii):

$$R \propto \frac{l}{A}$$

$$R = \frac{\rho l}{A}$$

$$\rho = \frac{RA}{l}$$

Where

ρ = Specific resistance (or) resistivity

Unit: Ohm meter

A = Area perpendicular to the current flow

Note:

'R' also depends upon the temperature, and the dependency is given as:

Material	Temperature	Resistance	Temperature Coefficient
Conductors	↑	↑	Positive
Semi-conductors	↑	↓	Negative
Insulators	↑	↓	Negative

Table 2.1

$R_2 = R_1[1 + \alpha_1 (T_2 - T_1)]$

Where

α_1 = Temperature coefficient (/°C)

R_1 = Initial resistance (ohm)

R_2 = Final resistance (ohm)

T_1 = Initial temperature (°C)

T_2 = Final temperature (°C)

Example 1:

A conductor has a length 'l' with a uniform cross-sectional area 'A' and resistance 'R'. If conductor length is uniformly stretched to 2l, then what will be its effect on the value of resistance?

Sol:

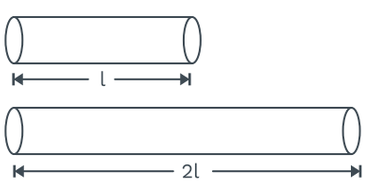


Fig. 2.8

$l_1 = l$ and $l_2 = 2l$

Volume (v_1) = Volume (v_2)

$A_1 * l = A_2 * (2l)$

$$A_2 = \frac{A_1}{2}$$

Note:

When the length of a wire is increased, then the cross-section area is reduced in the same amount.

$$R_1 = \frac{\rho_1 l_1}{A_1} = \frac{\rho_1 l}{A_1} \quad \text{... (i)}$$

{$\because \rho_1 = \rho_2$ for same material}

$$R_2 = \frac{\rho_2 l_2}{A_2} \quad \text{... (ii)}$$

$$R_2 = \frac{\rho_2 (2l)}{A_1/2} = 4.\frac{\rho_1 . l}{A_1}$$

$$R_2 = 4R_1$$

Hence, resistance becomes four times.

Note:
Resistance depends upon the type of material and its dimensions, but resistivity depends only on the type of material.

- **Conductance (G):** Reciprocal of resistance is called conductance. The symbol used to denote the conductance is 'G'.

  Where $G = \frac{1}{R} = \frac{I}{V}$

  $$\text{Conductance}(G) = \frac{1}{\text{Resistance}(R)} = \frac{\text{Current}(I)}{\text{Voltage}(V)}$$

  Unit $\rightarrow$ Ampere/Volt or Siemens or mho $(\mho)$ or $(\text{Ohm})^{-1}$
  Siemens is denoted by S.

- **Conductivity (σ):** Reciprocal resistivity is known as conductivity. The symbol used to denote the conductivity is 'σ'.
  where

  $$\sigma = \frac{1}{\rho} = \frac{1}{\frac{RA}{l}} = \frac{l}{RA}$$

  $$\text{Conductivity}(\sigma) = \frac{1}{\text{Resistivity}(\rho)} = \frac{1}{\text{ohm} - \text{meter}}$$

  Unit: $\text{Ohm}^{-1} - \text{m}^{-1}$ or Siemens/meter

Definitions

Conductance: It is defined as the reciprocal of resistance. It is denoted G conductance measures the inducement that is offered to flow the current through it. Unit of conductance is mho $(\mho)$ or Siemens (S).

Sign convention:

- The direction of current in voltage is an important consideration in Ohm's law.
- The polarity of voltage (V) and direction of current (I) should satisfy the sign convention.
- $V = IR \Rightarrow$ current is flowing from a higher potential to a lower potential.

- $V = -IR \Rightarrow$ current is flowing from a lower potential to a higher potential.

Concept of short-circuit and open-circuit:

Since the value of resistance may vary from 0 to ∞.

- If the resistance of any path is zero (R = 0), then that path can replace by a short-circuit.
- If the resistance of any path is infinity ($R = \infty$), then that path can replace by an open-circuit.
- The condition at which R = 0 is called short-circuit, as shown in figure 2.9 a
- For a short-circuit, $V = IR = 0 \Rightarrow$ Voltage is 0.
- For an open-circuit, $I = \frac{V}{R} = 0 \Rightarrow$ Current is 0.

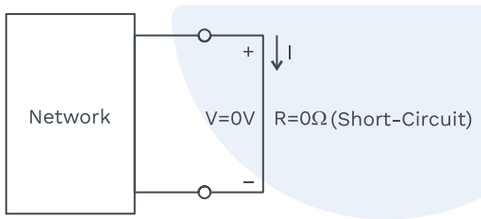


Fig. 2.9 a) Short-Circuit (R = 0)

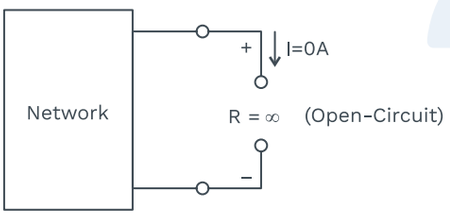


Fig. 2.9 b) Open-Circuit (R = ∞)

- Similarly, the condition at which $R = \infty$ is known as an open-circuit, as shown in figure 2.9 b

Grey Matter Alert!

Open-circuit: An open-circuit is a condition between two points in a circuit during which the resistance approaches infinity.

Short-circuit: In a short-circuit, the resistance between two points approaches zero.

Keep in mind:

- The voltage across the terminals of a short-circuit is always zero, current could be any finite value.
- The current through an open zero circuit is always zero; the voltage could be any finite value.

POWER AND ENERGY CONSUMED BY A RESISTOR

Power absorbed by a resistor:

The power can be expressed as:

$$P = V \cdot I = \frac{V^2}{R} = I^2R$$

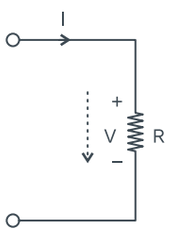


Fig. 2.10 Power Absorbed in a Resistor

Where
V = Voltage across a resistor.
I = current through a resistor.

We should note two things:

I) From the power expression, we can say that dissipated power in a resistor is a non-linear function of voltage/current.

II) Power dissipation in a resistor is always positive, which means the resistor absorbs power from the circuit. A resistor cannot generate energy at all. So, it is called a passive element.

Energy consumed by the resistor:

Energy is given as:

$$W = \int P \cdot dt \quad \text{or} \quad W = Pt$$

$$W = \int V \cdot I dt \quad \text{or} \quad W = V \cdot I \cdot t$$

$$W = \int \frac{V^2}{R} dt \quad \text{or} \quad W = \frac{V^2}{R} \cdot t$$

$$W = \int I^2 R \, dt \text{ Joules} \quad \text{or} \quad W = I^2Rt \text{ Joules}$$

Limitations of Ohm's law

- Ohm's law is applicable only for linear elements and the temperature must be constant.

Definitions

Energy: It is the ability to do the work, It is measured in Joules (J). Energy is measured in watt-hours (Wh) by electric power utilities in practice.

- Ohm's law is not applicable for semi-conductors, electrolytes, gaseous particles, Electric Iron etc.
- It is applicable only in the active region for semi-conductors.
- Ohm's law is not applicable to find out the short-circuit current.

$$\text{Slope} = \frac{I}{V} = \frac{I}{0} = \infty = \frac{1}{R}$$

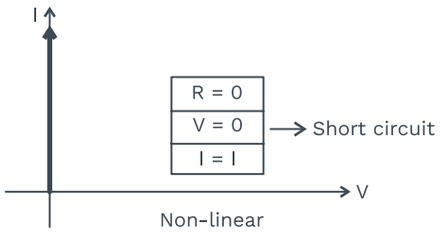
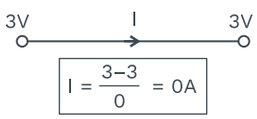


Fig. 2.11 Short Circuit Characteristics

a) **Non-linear:** Current is increasing, but voltage is not increasing.

b)

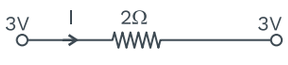


$$I = \frac{3-3}{0} = 0A$$

[Wrong calculation]

Fig. 2.12

c)



$$I = \frac{3-3}{2}$$

$$I = 0A$$

[Right calculation]

Fig. 2.13

Rack Your Brain

Is Ohm's law applicable for a non-linear circuit?

Note:

The above graph indicates the characteristics of a short-circuit. This graph is non-linear, so Ohm's law is not applicable to find the short-circuit current, and hence we use KCL to find the short-circuit current.

Ohm's law is not applicable to find the open-circuit voltage.

$$\text{Slope} = \frac{I}{V} = \frac{0}{V} = 0 = \frac{1}{R}$$

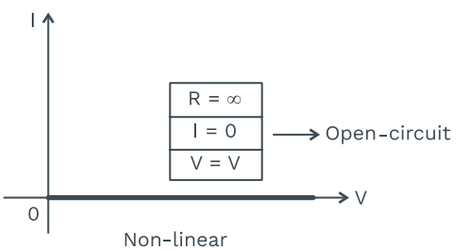
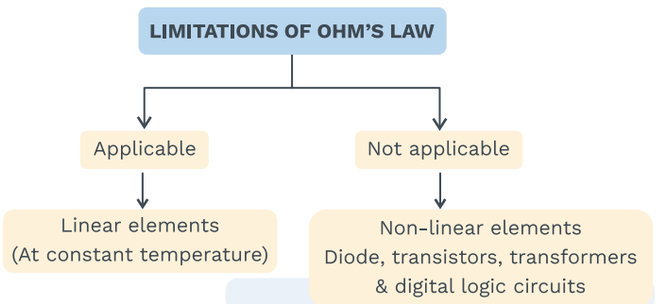


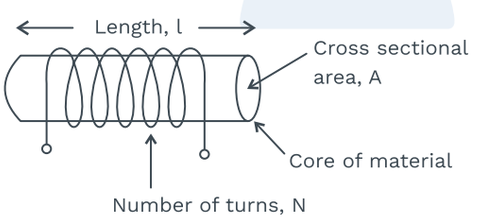
Fig. 2.14 Open circuit characteristics

Note:
The above graph indicates the characteristics of an open-circuit. This graph is non-linear, so Ohm's law is not applicable to find the open-circuit voltage, and hence we use KVL to find the open-circuit voltage.

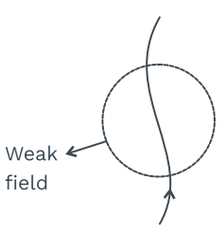


INDUCTOR AND INDUCTANCE

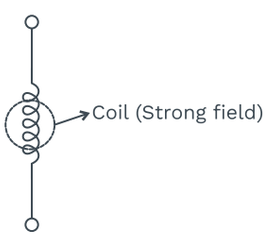
- An inductor is a coil or electromagnetic device that opposes any change in current. They consist of wire wound on an insulator.
- When the current or magnetic field changes, then e.m.f. is induced in the conductor. The rate of change of current is directly proportional to the induced emf.
- A cylindrical coil with a number of turns is called an inductor, as shown in figure 2.15 a below.



a) Typical form of an inductor



b) Wire (Weak field)



c) Strong magnetic field

Fig. 2.15 Different Forms of an Inductor

Definitions

Inductor: Inductance is the property of an inductor that causes it to resist the rate of change of current passing through it. It is measured in Henry (H).



Fig. 2.16

- Magnetic field lines are shown in fig. 2.17 a).

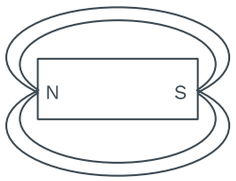


Fig. 2.17 a) Magnetic Field Lines

- Electric field lines are shown in figure 2.17 b).

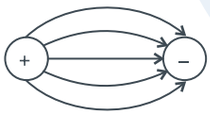


Fig. 2.17 b) Electric Field Lines

Rack Your Brain

Does an ideal inductor dissipate energy?

$N\varphi \propto I$

$N\varphi = LI$ (L is proportionality constant)

$$L = \frac{N\phi}{I} \frac{\text{Weber}}{\text{Ampere}} \text{ or Henry}$$

- A straight conductor, when twisted into a coil, forms a basic inductor.
- When a current flows through the inductor, it produces a magnetic flux (φ).
- The magnetic flux linkage (φ) is proportional to the current through the inductor.
- If the inductance does not depend upon the flux linkages or current, then it is said to be a linear inductor; otherwise non-linear.

Linear inductor: Air core inductor
Non-linear inductor: Iron core inductor

Inductance of a solenoid:
When an AC current flows through the conductor of a solenoid, it will create a varying magnetic field within the solenoid.
This variation in magnetic field also creates a variation in magnetic flux which will induce an emf inside the loops of the conductor.

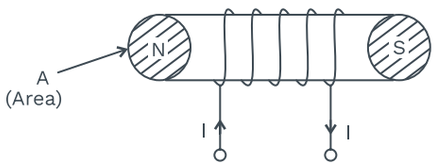


Fig. 2.18 Solenoid

$\rightarrow L \propto N^2$ ... (i)

$\rightarrow L \propto A$ ... (ii)

$\rightarrow L \propto \frac{1}{l}$ ... (iii)

Combining all the above equations

$$L \propto \frac{N^2 A}{l}$$

$$L = \frac{\mu N^2 A}{l} \text{ Henry}$$

$\mu \rightarrow$ Permeability, where $\mu = \mu_0 \mu_r$

μ_0 = Absolute permeability

$= 4\pi \times 10^{-7}$ Henry/meter

μ_r = Relative permeability (unitless)

$\mu_0 = 1$ for air

We know-

$$H = \frac{NI}{l} \text{ and } B = \mu H$$

$$B = \frac{\mu NI}{l}$$

$$\frac{\phi}{A} = \frac{\mu NI}{l}$$

$$\phi = \frac{A\mu NI}{l}$$

$$\frac{LI}{N} = \frac{\mu ANI}{l} \quad \left\{\because \phi = \frac{LI}{N}\right\}$$

$$L = \frac{N^2 \mu A}{l} \text{ Henry}$$

Rack Your Brain

Why an inductor behaves like a short-circuit initially when the supply is DC?

Faraday's laws:

- **First law:** Whenever a conductor cuts the rate of change of flux, an emf is induced in that conductor.

$$emf = -N\frac{d\phi}{dt} \quad \text{(Lenz's law)}$$

- **Second law:** The rate of change of flux determines the induced emf in a conductor.

$$v \propto \frac{d\phi}{dt}$$

Voltage, current and energy relationship of an inductor:

The voltage across an inductor is related to the rate of change in current with respect to time.

$$v = L\frac{di}{dt} \qquad \ldots (i)$$

Where

L = inductance of the inductor.

- It is measured in Henry (H).

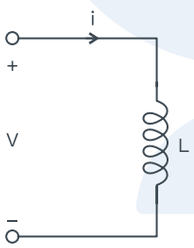


Fig. 2.19 Circuit Symbol for an Inductor

- An inductor whose inductance is independent of the current is known as a linear inductor.

The current-voltage relationship from equation i) can be given as:

$$di = \frac{1}{L}vdt$$

$$i = \frac{1}{L}\int_{-\infty}^{t} vdt \qquad \ldots (ii)$$

$$\text{Or } i = \frac{1}{L}\int_{t_0}^{t} vdt + i(t_0) \qquad \ldots (iii)$$

Where $i(t_0)$ is the current that accumulates from $t = -\infty$ to t_0. Equation iii) shows that the inductor current also depends on the past value of the voltages. Hence, we can say that an inductor has some memory. The power delivered to an inductor is:

$$P = vi = \left(L\frac{di}{dt}\right)i$$

The energy stored is, $W = \int_{-\infty}^{t} Pdt$

$$W = \int_{-\infty}^{t}\left(L\frac{di}{dt}\right)idt$$

$$W = \frac{1}{2}Li^2(t) - \frac{1}{2}Li^2(-\infty)$$

Since, $i(-\infty) = 0$

$$\therefore \quad W = \frac{1}{2}Li^2$$

Rack Your Brain

Why does an inductor not allow sudden changes in the current?

Note:

Even though the instantaneous power, p(t) may be positive or negative, the total energy stored in the inductor $\left(\frac{1}{2}Li^2\right)$ is always positive or zero.

Inductor combinations:

- **Series inductor:**

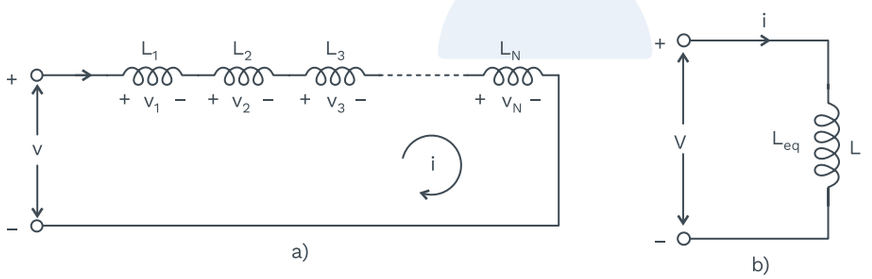


Fig. 2.20 a) Series Combination of N Inductors
b) Equivalent Circuit of Series Inductors

Let us take the case of N number of inductances (L_1, L_2, L_3, ... L_N) connected in series as shown in figure 2.20, and the combination is connected to a voltage source v where v_1, v_2, ... v_N be the voltage drops across respective inductances. Applying KVL in the loop of figure 2.20 a).

Substituting $v_N = L_N\frac{di}{dt}$ results in:

Grey Matter Alert!

Inductor: If current is constant, then there is no voltage difference across the inductor, i.e. if the current is not changing with time, the inductor behaves like a short-circuit.

$$v = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + ... + L_N \frac{di}{dt}$$

$$v = \left(L_1 + L_2 + ... + L_N\right) \frac{di}{dt}$$

$$v = L_{eq} \frac{di}{dt}$$

and $L_{eq} = L_1 + L_2 + L_3 + ... + L_N$

Inductances in series add to give the equivalent inductance, i.e. equivalent inductance is the sum of individual inductances.

Definitions

Series inductor: Inductances in series add to give the equivalent inductance, i.e. equivalent inductance is the sum of individual inductances.

- **Parallel inductor:**

Now let us take the case of a parallel combination of N inductances L_1, L_2, ... L_N as shown in the figure 2.21 a) and b). We know that in a parallel combination of elements, voltage across them is always the same.

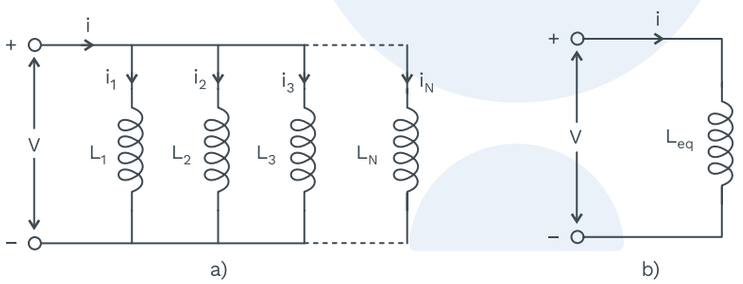


Fig. 2.21 a) Parallel Combination of N Inductors
b) Equivalent Circuit of Parallel Inductors

By applying KCL, we have:

$$i = i_1 + i_2 + i_3 + ... + i_N$$

But, $i_N = \frac{1}{L_N} \int_{-\infty}^{t} v\,dt$

$$i = \frac{1}{L_1} \int_{-\infty}^{t} v\,dt + \frac{1}{L_2} \int_{-\infty}^{t} v\,dt + ... + \frac{1}{L_N} \int_{-\infty}^{t} v\,dt$$

$$i = \left(\frac{1}{L_1} + \frac{1}{L_2} + ... + \frac{1}{L_N}\right) \int_{-\infty}^{t} v\,dt$$

$$i = \frac{1}{L_{eq}} \int_{-\infty}^{t} v\,dt$$

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + ... + \frac{1}{L_N}$$

Grey Matter Alert!

The current flowing through an inductor must be a continuous function of time; there cannot be any abrupt fluctuations in current in the inductor. Sudden changes in current require infinite voltage in the inductor.

- The equivalent inductance of parallel inductors is reciprocal to the sum of the reciprocals of the individual inductances.
- Thus, we can say that the reciprocal of inductances in parallel add to give the reciprocal of equivalent inductance, i.e. reciprocal of equivalent inductances, is the sum of reciprocals of individual inductances.

Example 2:
An electric Iron (Press) draws 2A at 240V. Find its resistance.
Sol:
This question is based on Ohm's law. From Ohm's law:

$$R = \frac{V}{I}$$

$$R = \frac{240}{2}$$

$$R = 120\Omega$$

Hence, the resistance is 120Ω.

Example 3:
For a circuit, if a voltage source of $20\sin\pi t$ is applied across a 10kΩ resistor. Evaluate the current and power dissipation through the resistor.

Sol:

$$i = \frac{V}{R} = \frac{20\sin\pi t}{10k} = 2\sin\pi t\,mA$$

$$\text{Power dissipated} = \frac{(V_{rms})^2}{R} = \frac{\left(20/\sqrt{2}\right)^2}{10000} = 20\,mW$$

Example 4:
Calculate the current through an inductor of value 5H, if the voltage across it is given as:

$$V(t) = \begin{cases} 30t^2 & , \quad t > 0 \\ 0 & , \quad t < 0 \end{cases}$$

Also, find the energy stored at t = 5 sec.

Sol:

Since, $i = \frac{1}{L}\int_{t_0}^{t} V(t)\,dt + i(t_0)$ & L = 5H

$$i = \frac{1}{5}\int_{0}^{t} 30t^2 dt + 0 = 6 \times \frac{t^3}{3} = 2t^3 A$$

Definitions

Parallel inductor: The reciprocal of the sum of the reciprocals of individual inductances give the equivalent inductance of the parallel combination of inductors.

The power $P = Vi = 60t^5$, and the energy stored is,

$$W = \int Pdt = \int_0^5 60t^5 dt = 60\frac{t^6}{6}\Big|_0^5$$

$$W = 156.25 \text{ kJ}$$

Example 5:

For the given circuit, calculate the equivalent inductance.

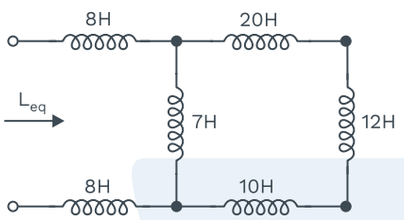
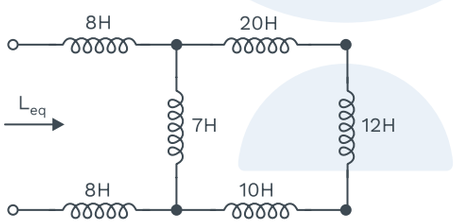


Fig. 2.22

Sol:



Here, 10H, 12H and 20H are in series combination, so we get 42H equivalent inductance of them.

And, this 42H is parallel with 7H, so the equivalent inductance of them is

$$= \frac{7 \times 42}{7 + 42} = 6\text{H}$$

Now, 6H is in series with 8H and 8H. Hence the equivalent inductance is:

$L_{eq} = 8 + 6 + 8 = 22\text{H}$

$L_{eq} = 22\text{H}$

CAPACITORS AND CAPACITANCE:

- Capacitor is widely used in various electronic circuits.
- A capacitor having two parallel conducting plates, separated by a dielectric material.
- Capacitance may be defined as the amount of charge required to develop a 1-volt potential difference between its plates, and the unit of this capacitance is as Faraday.

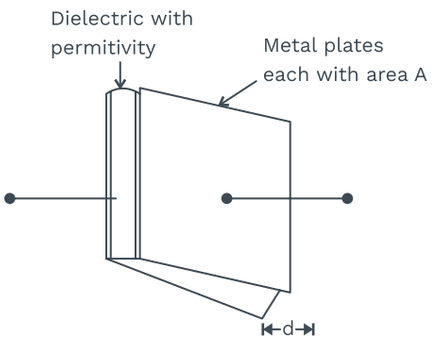


Fig. 2.23

If a voltage source V is applied to the capacitor, as in figure 2.24, then a positive charge Q will reflect on one plate and a negative charge –Q on the other plate.

The amount of charge stored, represented by Q, is directly proportional to the applied voltage V so that:

$$Q = CV$$

Where C = capacitance of the capacitor.

- It is measured in Farad (F).
- Capacitance is a measure of the ability of the device to store energy in the form of a separated charge or an electric field.
- The capacitance of a parallel plate capacitor is given by:

$$C = \frac{\in A}{d} \text{ Farad}$$

Where

$\in$ = the dielectric constant,
A = the area of the plates, and
d = the spacing between plates.

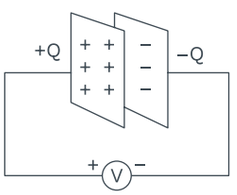


Fig. 2.24 Parallel Plate Capacitor

Keep in mind:

An inductor and capacitor stores release energy (i.e. have a memory) but a resistor always dissipates energy.

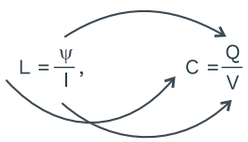


Fig. 2.25

- If the capacitance of a capacitor does not depend upon the charge or voltage, then it is said to be a linear capacitor; otherwise non-linear.

Capacitance of capacitor

$C \propto A$ ... (i)

$C \propto \frac{1}{d}$ ... (ii)

Combining $C \propto \frac{A}{d}$

$$C = \frac{\in A}{d}$$

Where

$\in$ = permittivity (Farad/meter)

$\in = \in_0 \in_r$

$\in_0$ = Absolute permittivity = 8.854×10^{-12} F/m

$\in_r$ = Relative permittivity (Unitless)

$\in_r$ = 1 for air. (Unitless)

Definitions

Capacitance: Capacitance is the ratio of the charge on one plate of a capacitor to the voltage difference between the two plates. It is measured in Farad (F).

Voltage, current and energy relationship of a capacitor:

To obtain the current voltage relationship of the capacitor, we use the equation-

$i = \frac{dq}{dt}$ ... (i)

Putting q = CV in equation i), we get:

$$i = \frac{dq}{dt} = C\frac{dV}{dt}$$

$dV = \frac{1}{C} i\,dt$ ... (ii)

Where 'V' and 'i' follow the passive sign convention as shown in figure 2.12. We can see that the voltage-current relation of a capacitor is a linear relation.

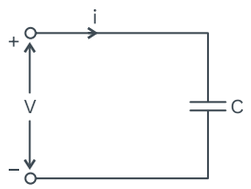


Fig. 2.26 Circuit Symbol of a Capacitor

- **Voltage across capacitor:**

The voltage-current relationship of a capacitor can be obtained by integrating both sides of equation ii). On integrating, we get:

$$V(t) = \frac{1}{C}\int_{-\infty}^{t} i(t)\,dt \qquad \text{... (iii)}$$

$$V(t) = \frac{1}{C}\int_{-\infty}^{0} i(t)dt + \frac{1}{C}\int_{0}^{t} i(t)dt$$

$$V(t) = V(0) + \frac{1}{C}\int_{0}^{t} i(t)\,dt$$

- Reciprocal of capacitance is 'ELASTANCE'
  Unit: DARAF

Instantaneous power:

The instantaneous power delivered to a capacitor is

$$P = V(t)i(t) = CV(t)\frac{dV(t)}{dt} \qquad \text{... (iv)}$$

Energy stored in a capacitor:

Energy stored in a capacitor is, therefore:

$$W = \int_{-\infty}^{t} P(t)\,dt$$

$$W(t) = \int_{-\infty}^{t} V(t)\cdot C\frac{dV(t)}{dt}\cdot dt$$

$$W(t) = C\int_{-\infty}^{t} V(t)\cdot dV(t)$$

$$W(t) = \frac{1}{2}C\left|(V(t))^2\right|_{-\infty}^{t}$$

$$W(t) = \frac{1}{2}C\left\{(V(t))^2 - (V(-\infty))^2\right\}$$

Rack Your Brain

Does an ideal capacitor dissipate power?

$V(-\infty) = 0$

$$W(t) = \frac{1}{2} C \left[V(t) \right]^2$$

$$W(t) = \frac{1}{2} q(t) \cdot V(t)$$

$$W(t) = \frac{1}{2} \frac{q^2(t)}{C}$$

Or

$$W(t) = \frac{1}{2} CV^2 \text{ Joule}$$

- Energy stored in a capacitor depends upon the instantaneous value of voltage across the capacitor.
- Energy stored in a capacitor is in the form of an electric field, i.e. potential energy.

  $V \rightarrow$ electric field

  $I \rightarrow$ magnetic field

Grey Matter Alert!

If the current is not changing with time (DC), the capacitor behaves as an open-circuit and the voltage on a capacitor cannot change suddenly.

PROPERTIES OF A CAPACITOR

1) A capacitor does not allow the sudden change in voltage for all practical inputs.

2) At the time of switching, the capacitor acts as a short-circuit.

3) When a dc voltage is connected to the capacitor, the capacitor behaves like an open-circuit in a steady-state.

For Example:

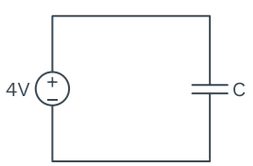


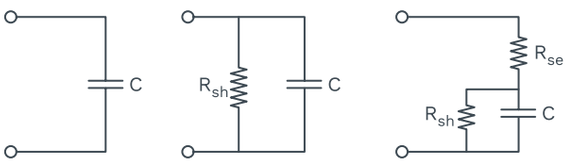
Fig. 2.27

$$i = C \frac{dV}{dt}$$

$$i = C \frac{d}{dt}(4)$$

$i = 0$ (open-circuit)

4) An ideal capacitor does not dissipate the energy.



(a) Ideal (b) Practical Capacitance (c) Exact model

Fig. 2.28 Capacitor Models

- The shunt resistance R_{sh} is very high (in the exact model).
- Intensity of the spark indicates the strength of the capacitor.
- If the supply is disconnected, then only the capacitor stores energy. The inductor does not store any energy.

Keep in mind:

We see that even though the instantaneous power p(t) may be positive or negative, and the total energy stored in a capacitor is given by the equation $W = \frac{1}{2}CV^2$ is always positive.

Rack Your Brain

Why a capacitor does not allow sudden change in voltage?

CAPACITOR COMBINATIONS

Series capacitors: The equivalent capacitance (C_{eq}) of N capacitors connected in series can be obtained by comparing the circuit in figure 2.29 a) with the equivalent circuit in figure 2.29 b). Note that the same current 'i' flows through each capacitor.

Applying KVL to the loop in figure 2.29 a).

$$V = V_1 + V_2 + V_3 + ... + V_N$$

But, $V_N = \frac{1}{C_N}\int_{-\infty}^{t} i\,dt$

$$V = \frac{1}{C_1}\int_{-\infty}^{t} i\,dt + \frac{1}{C_2}\int_{-\infty}^{t} i\,dt + ... + \frac{1}{C_N}\int_{-\infty}^{t} i\,dt$$

$$V = \left(\frac{1}{C_1} + \frac{1}{C_2} + ... + \frac{1}{C_N}\right)\int_{-\infty}^{t} i\,dt = \frac{1}{C_{eq}}\int_{-\infty}^{t} i\,dt$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + ... + \frac{1}{C_N}$$

Definitions

Series capacitance: The reciprocal of the resultant or equivalent capacitance is equal to the sum of the reciprocals of the individual capacitances.

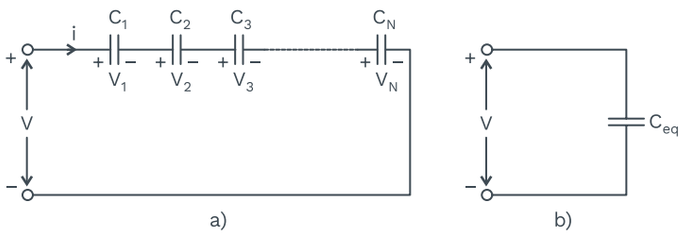


Fig. 2.29 a) Series Combination of N Capacitors
b) Equivalent Circuit of Series Capacitors

Parallel capacitor:

In order to obtain the equivalent capacitance C_{eq} of N capacitors in parallel, consider the circuit in figure 2.30 a) and the equivalent circuit is in figure 2.30 b).

In the parallel form, the voltage across all the capacitors is the same.

Apply KCL, we get:

$$i = i_1 + i_2 + i_3 + ... + i_N$$

$$i_N = C_N \frac{dV}{dt}$$

$$i_N = C_1 \frac{dV}{dt} + C_2 \frac{dV}{dt} + ... + C_N \frac{dV}{dt} = C_{eq} \frac{dV}{dt}$$

$$C_{eq} = C_1 + C_2 + C_3 + ... + C_N$$

The equivalent capacitance of N parallel-connected capacitors is the sum of the individual capacitances.

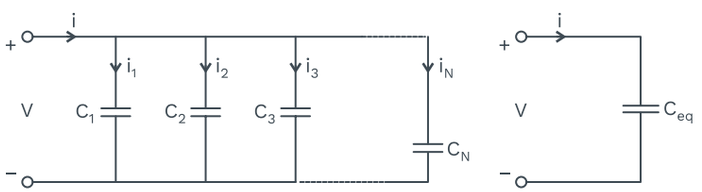


Fig. 2.30 a) Parallel Combination of N Capacitors
b) Equivalent Circuit of Parallel Capacitors

Example 6:

Find the current through a capacitor of 200μF; if a voltage is applied across it, gives in the figure below:

Rack Your Brain

Why does the capacitor behave as an open circuit for DC voltages?

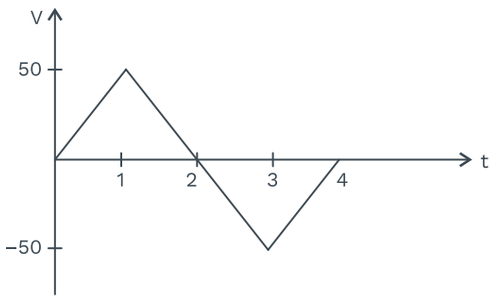
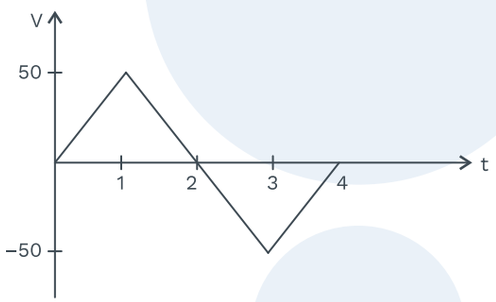


Fig. 2.31

Sol:

Given,

The voltage is shown below:



Mathematically, It is written as:

$$V(t) = \begin{cases} 50t\,V & 0 \le t < 1 \\ (100 - 50t)\,V & 1 \le t < 3 \\ (-200 + 50t)\,V & 3 \le t < 4 \\ 0 & \text{otherwise} \end{cases}$$

Since, $i = C\dfrac{dV}{dt}$ and C = 200 μF,

$$i(t) = 200 \times 10^{-6} \times \begin{cases} 50 & 0 \le t < 1 \\ -50 & 1 \le t < 3 \\ 50 & 3 \le t < 4 \\ 0 & \text{otherwise} \end{cases}$$

$$i(t) = \begin{cases} 10\,mA & 0 \le t < 1 \\ -10\,mA & 1 \le t < 3 \\ 10\,mA & 3 \le t < 4 \\ 0 & \text{otherwise} \end{cases}$$

Definitions

Parallel capacitor: The resultant or equivalent capacitance is equal to the sum of the individual capacitances. It is to be noted that the capacitors in parallel combine in the same way as resistors in series.

Now, from the above expression of i(t), we can draw the current waveform, which is given below:

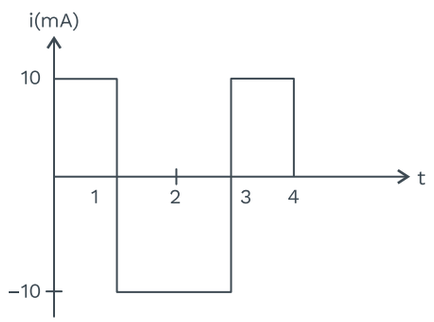


Fig. 2.32

Previous Years' Question

The impedance of a parallel circuit is (10 – j30) Ω at 1 MHz. The values of circuit elements will be

a) 10Ω and 6.4 mH
b) 100Ω and 4.7 nF
c) 10Ω and 4.7 mH
d) 100Ω and 6.4 nF
Sol: b)

Example 7:
For the network given below, what will be the equivalent capacitance (C_{eq}) between A and B?

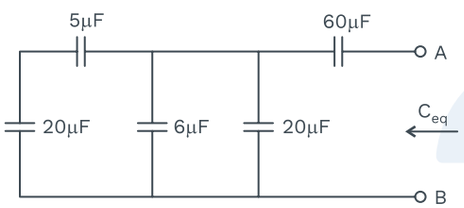
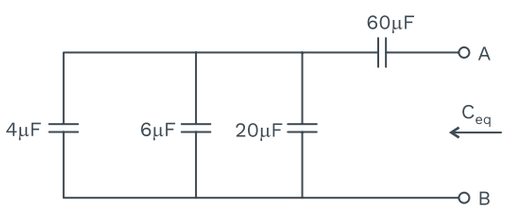


Fig. 2.33

Sol:
The 20μF and 5μF capacitors are in series. Their equivalent capacitance is:

$$\frac{20 \times 5}{20 + 5} = \frac{100}{25} = 4\mu F$$



Now, 4μF is parallel with 6μF and 20μF.
It's equivalent capacitance = 4 + 6 + 20 = 30μF

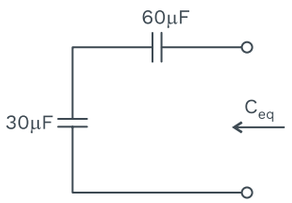


Fig. 2.34

Now, 30μF is a series with 60μF.
So, the equivalent capacitance for the entire circuit is

$$C_{eq} = \frac{30 \times 60}{30 + 60} = 20\mu F$$

$$C_{eq} = 20\mu F$$

NODES, BRANCHES AND LOOPS:

- **Branch:** A branch of a network is defined as a group of elements connected in series or parallel combinations and which has two terminals from connections.

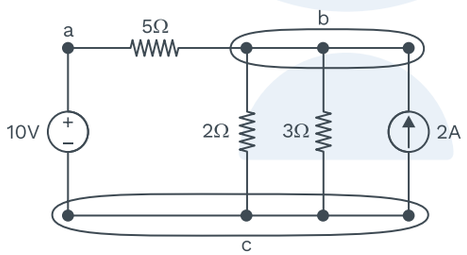


Fig. 2.35 A Typical Network

- For the above network:
  Number of branches = 5
  Branch 1 ⇒ 10V voltage source
  Branch 2 ⇒ 5Ω resistance
  Branch 3 ⇒ 2Ω resistance
  Branch 4 ⇒ 3Ω resistance
  Branch 5 ⇒ 2A current source

Node: A node is a common point in a network at which two or more branches joint.

- In figure 2.15:
  Number of nodes = 3
  Node 1 ⇒ a
  Node 2 ⇒ b
  Node 2 ⇒ c

- It is basically a connection point between two or more elements.
- The above network is redrawn as in figure 2.16, which clearly represents three nodes a, b and c.

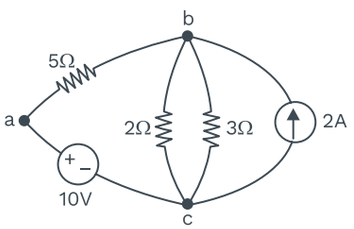


Fig. 2.36 Simplified Network Of Figure 2.35

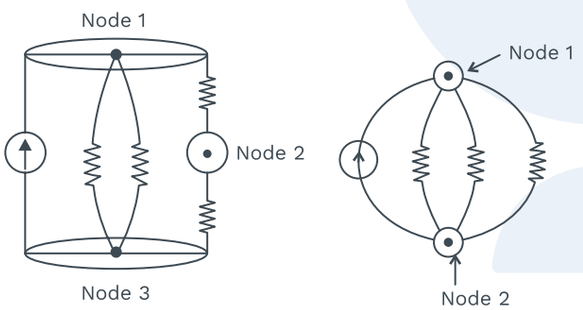


Fig. 2.37 a) 3 Nodes, 5 Branches
b) 2 Nodes, 4 Branches

Loop: A loop is a set of branches which forms a closed path in a network.

- A loop is said to be independent if it contains at least one branch, which is not a part of any other independent loop.
- If a loop consists of at least one branch, which is a not part of any other loop, then the loop is called an independent loop.
- A network with b branches, n nodes and l independent loops will satisfy the fundamental theorem of network topology.

$$b = l + n - 1$$

Previous Years' Question

For the circuit shown in the figure, the capacitance measured between terminals, B and Y will be:

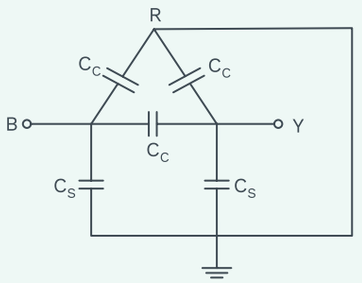


Fig. 2.38

a) $C_C + \left(\frac{C_S}{2}\right)$

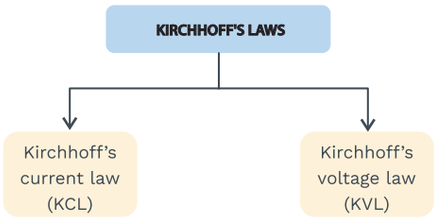
b) $C_S + \left(\frac{C_C}{2}\right)$

c) $\frac{(C_S + 3C_C)}{2}$

d) $3C_C + 2C_S$

Sol: c)

KIRCHHOFF'S LAWS



Kirchhoff's current law:
For a lumped network, the algebraic sum of currents entering or leaving a node is equal to zero at every instant of time.

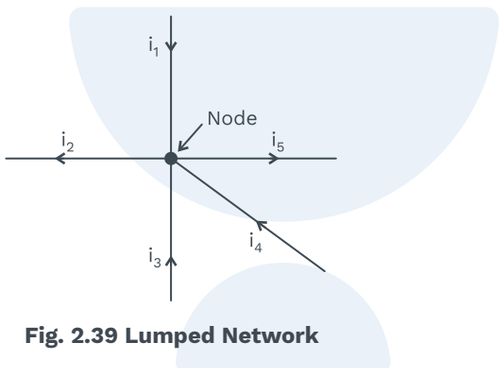


Fig. 2.39 Lumped Network

$$\sum_{n=1}^{N} i_n = 0$$

Consider the node in figure 2.39 Applying KCL:

$$i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0 \quad \text{... (i)}$$

By rearranging the terms, we get-

$$i_1 + i_3 + i_4 = i_2 + i_5 \quad \text{... (ii)}$$

Equation ii) is an alternative form of KCL.

- Kirchhoff's current law is based on the law of conservation of charge, which states that the algebraic sum of charges within a system cannot change.
- By this law, currents entering a node may be positive, while currents leaving a node may be taken as negative or vice versa.
- Sum of the currents entering the node is equal to the sum of currents leaving the node.

$$\frac{q_1}{t} + \frac{q_3}{t} + \frac{q_4}{t} = \frac{q_2}{t} + \frac{q_5}{t}$$ (From figure 2.39)

- Sum of charges entering the node is equal to the sum of the charges leaving the node, i.e. no charge is accumulating.

$n_1e + n_3e + n_4e = n_2e + n_5e$ (From figure 2.39)

$n_1 + n_3 + n_4 = n_2 + n_5$

Keep in mind:

We are not sure of the direction of current through a particular element within the circuit. In such cases, we assume a reference direction and perform all the calculations on this assumption. If our assumption is incorrect, calculations will show that the current has a negative sign. The negative sign simply indicates that the actual current is in the opposite direction from the selected reference direction.

KCL for a closed surface:

For a lumped network, the algebraic sum of current, entering or leaving a closed surface is equal to zero at every instant of time.

[Closed surface → Gaussian surface]

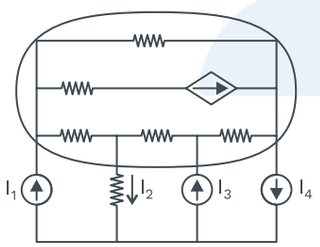


Fig. 2.40 KCL Applied to a Closed Surface

In general, KCL applies to any closed surface or boundary. Since a node may be regarded as a closed surface shrunk to a point. In two dimensions, a closed boundary is the same as a closed path. As an example, consider the circuit of figure 2.40.

- If we define a closed surface as shown in figure 2.40 and apply KCL, we will get:

$$I_1 + I_3 = I_2 + I_4$$

Example 8:
In the figure shown below, find the current 'I'

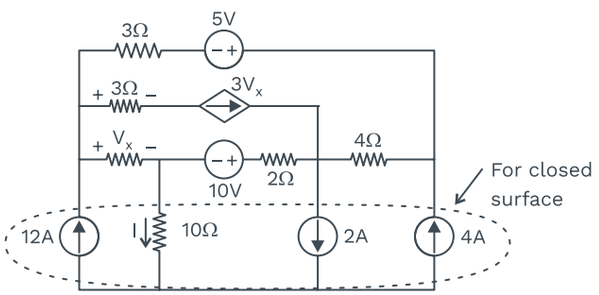


Fig. 2.41

Sol:
Applying KCL, we get

$12 + 4 = 2 + I$

$I = 14$ A

Example 9:
In the figure shown below, find the current 'I'

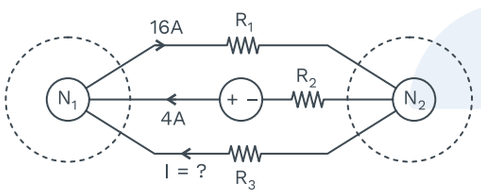


Fig. 2.42

Sol:
Applying KCL,

$4 + I = 16$

$I = 12$ A

Definitions

KCL: Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

Features:

- Kirchhoff's current law is applicable to any lumped network, irrespective of the type of network elements such as linear, non-linear, time-variant, time-invariant, unilateral, bilateral, active and passive, i.e. KCL is independent of the nature of the elements.
- KCL is based on the conservation of charge, i.e. charge can neither be created nor destroyed.
- KCL is also applicable to any closed surface.
- KCL is not applicable to high frequency and distributed networks.

Example 10:
In the figure shown below, find the current 'I'

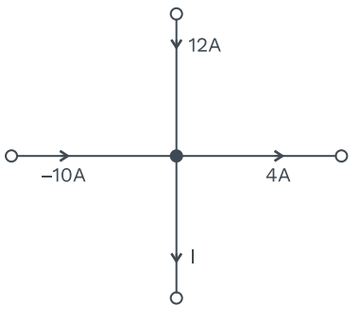


Fig. 2.43

Sol:
Applying KCL:

$-10 + 12 - 4 - I = 0$

$I = -2$ A

Example 11:
Find current I_y in the given circuit.

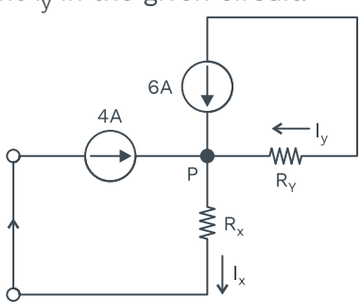
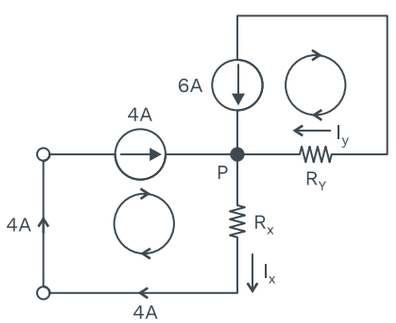


Fig. 2.44

Rack Your Brain

Are KCL and KVL valid for circuits with non-linear elements?

Sol:



Applying KCL at node P:

$4 + 6 + I_y - I_x = 0$

$I_x - I_y = 10A$

$I_x = 4A$

$I_y = -6A$

Note:

Two loop currents are independent of each other.

Example 12:

If $i_R(t) = 4e^{-3t} + 3e^{-4t}$ A, $i_L(0) = 1A$ then, find φ = ?

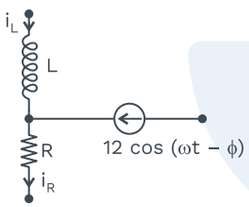


Fig. 2.45

Rack Your Brain

Can we apply KCL and KVL to AC circuits?

Sol:

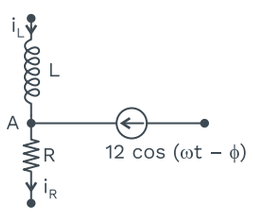
KCL at node A,

$12 \cos(\omega t - \varphi) + i_L(t) = i_R(t)$

At t = 0

$12 \cos(0 - \varphi) + i_L(0) = i_R(0)$

$12 \cos \varphi + 1 = i_R(0)$



$$\begin{cases} i_R(t) = 4e^{-3t} + 3e^{-4t} \\ i_R(0) = 4e^{0} + 3e^{0} \\ i_R(0) = 7A \end{cases}$$

$12 \cos \varphi + 1 = i_R(0) = 4e^0 + 3e^0$

$12 \cos \varphi = 4 + 3 - 1$

$$\cos\phi = \frac{6}{12} = \frac{1}{2}$$

$$\cos\phi = \frac{1}{2}$$

$$\phi = \cos^{-1}\left(\frac{1}{2}\right)$$

$$\phi = \frac{\pi}{3}$$

Kirchhoff's voltage law (KVL):

- Kirchhoff's voltage law is used to find the unknown currents in a given network.
- It states that the algebraic sum of all the voltages around a closed path in a circuit must be equal to zero. Mathematically:

$$\sum_{\text{loop}} \text{Voltage across each elements} = 0$$

- **Statement 1:** For a lumped network, the algebraic sum of voltage drops around any closed path is equal to zero.

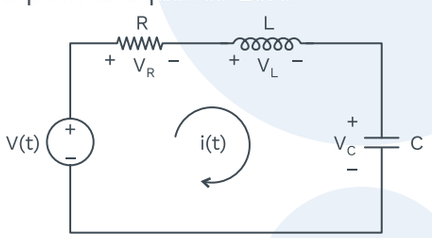


Fig. 2.46 RLC Circuit

Applying KVL,

$$V(t) - V_R - V_L - V_C = 0$$

$$V(t) = V_R + V_L + V_C$$

$$V(t) = R\,i(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)\,dt$$

- **Statement 2:** For a lumped network, the algebraic sum of node voltages for any closed node, sequence is equal to zero at every instant of time.

Example 13:
Determine V_1 and V_2, if $V_D = 0$.

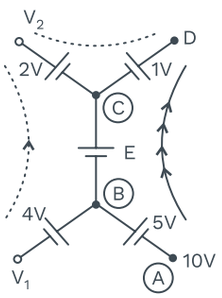


Fig. 2.47

Sol:
Closed node sequence, A-B-C-D:

$\Rightarrow$ $10 + 5 + E + 1 = 0$

$E = -16$ V

$\Rightarrow$ $V_1 + 4 - 5 - 10 = 0$

$V_1 = 11$ V

$\Rightarrow$ $V_2 - 2 + 1 = 0$

$V_2 = 1$ V

Features:
KVL is applicable to any lumped network, irrespective of the type of network elements such as linear, non-linear, time-variant, time-invariant unilateral, bilateral, active and passive, i.e. independent of the nature of elements.

- KVL is based on the conservation of energy.

  $V = V_R + V_C + V_L$ (Figure 2.20)

  $$\frac{W}{Q} = \frac{W_R}{Q} + \frac{W_C}{Q} + \frac{W_L}{Q}$$

  $W = W_R + W_C + W_L$
- KVL is also applicable to a closed node sequence.
- KVL is not applicable to high frequency and distributed networks.

Example 14:
For the circuit shown in the figure, find the value of the current.

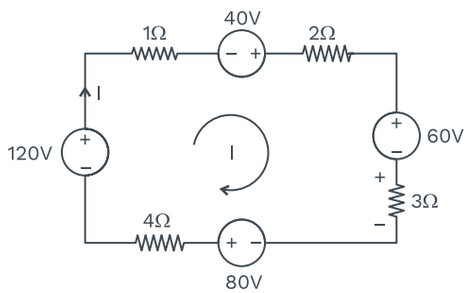
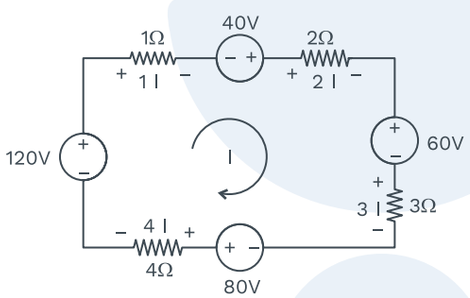


Fig. 2.49

Sol:



Applying KVL in the loop,

$120 - I + 40 - 2I - 60 - 3I + 80 - 4I = 0$

$180 - 10I = 0$

$I = 18A$

Or

$$I = \frac{\Sigma V}{\Sigma R}$$

$$I = \frac{120 + 40 - 60 + 80}{1 + 2 + 3 + 4} = \frac{180}{10}$$

$I = 18A$

Sign conventions:

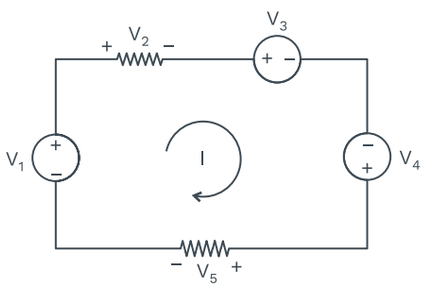


Fig. 2.50 Sign Convention

While moving around a loop in any direction (either clockwise or anti clockwise), a plus sign is assigned to a potential rise (– to +) and a minus sign to potential drop (+ to –). For example, consider the circuit of figure 2.50, where voltage V_1 and V_4 are taken as voltage rises. Applying KVL around clockwise direction, we get:

$$V_1 - V_2 - V_3 + V_4 - V_5 = 0$$

$$V_1 + V_4 = V_2 + V_3 + V_5$$

SERIES RESISTANCES AND VOLTAGE DIVISION:

Two or more circuit elements are said to be in series if they exclusively share a single node and consequently carry the same current.

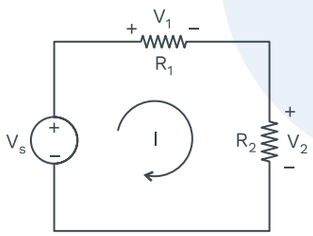


Fig. 2.51 Two-Series Connected Resistors

Consider the single loop circuit in figure 2.51. Two resistances R_1 and R_2 are connected in series and the same current I flows through each of them.

Applying KVL in the loop,

$V_s - V_1 - V_2 = 0$ ... (i)

Now, applying Ohm's law to each resistance,

$V_1 = iR_1$ ... (ii)

$V_2 = iR_2$ ... (iii)

Putting V_1 and V_2 in equation i),

$V_s - iR_1 - iR_2 = 0$

$V_s = i(R_1 + R_2)$ ... (iv)

Equation iv) can be written as,

$V_s = iR_{eq}$

Where R_{eq} is the equivalent resistance of two resistances connected in series, given,

$R_{eq} = R_1 + R_2$

Previous Years' Question

The voltage V in the figure is equal to:

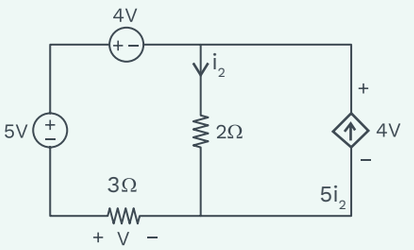


Fig. 2.52

a) 3V
b) –3V
c) 5V
d) None of these
Sol: d)

Keep in mind:

If two resistors with resistances R_1 and R_2 are connected in series, then their equivalent resistance is the sum of the individual resistances, and it is given by $R_{eq} = R_1 + R_2$.

Now, the voltage across each resistance can be obtained as:

$$V_1 = \frac{R_1}{R_1 + R_2} V_s \text{ and } V_2 = \frac{R_2}{R_1 + R_2} V_s$$

We can see that the source voltage V_s is divided among the resistors in direct proportion to their resistance values. This is called the principle of voltage division, and the circuit figure 2.52 is called voltage divider circuit.

Rack Your Brain

What are the conditions for which Kirchhoff's law is not valid?

Note:

In general, if a voltage divider circuit has N resistors with values R_1, R_2, R_3 ... R_N is series with the source voltage V_s, then the voltage across the Nth resistor (R_N) will be $V_N = \frac{R_N}{R_1 + R_2 + ... + R_N} V_s$.

PARALLEL RESISTANCES AND CURRENT DIVISION:

Two or more circuit elements are said to be in parallel if the same voltage appears across each of the elements.

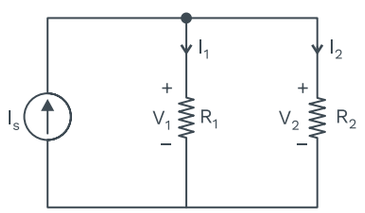


Fig. 2.53 Two Parallel Connected Resistors

Rack Your Brain

Why the current always remains same in a series circuit?

Consider the figure 2.53, where two resistors are connected in parallel and therefore same voltage appears them.

We write KCL at the top node,

$$I_s = I_1 + I_2 \quad \text{... (i)}$$

From ohm's law, $V = V_1 = V_2 = I_1R_1 = I_2R_2$

$$I_1 = \frac{V}{R_1} \quad \text{... (ii)}$$

$$I_2 = \frac{V}{R_2} \quad \text{... (iii)}$$

Substituting I_1 and I_2 into equation i)

$$I_s = \frac{V}{R_1} + \frac{V}{R_2} = V\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{V}{R_{eq}} \quad \text{... (iv)}$$

Where R_{eq} is the equivalent resistance of the resistors connected in parallel and it is given as:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} \quad \text{... (v)}$$

Note:

If N resistors with resistances R_1, R_2, R_3 ... R_N are connected in parallel, then the equivalent resistance given by:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + ... + \frac{1}{R_N}$$

- The equivalent resistance of any number of parallel-connected resistances is always less than the smallest value of resistance in the combination.

Note:

We obtain current through each of the resistance as follows:

From equation iv)

$$V = I_s R_{eq} = \left(\frac{R_1 R_2}{R_1 + R_2}\right) I_s \quad \text{... (vi)}$$

Substituting equation vi) into equations ii) and iii), we have,

$$I_1 = \left(\frac{R_2}{R_1 + R_2}\right) I_s \quad \text{... (vii)}$$

$$I_2 = \left(\frac{R_1}{R_1 + R_2}\right) I_s \quad \text{... (viii)}$$

Equation vii) and viii) implies that the total current I_s is distributed among the resistors in inverse proportion to their resistance values. This is known as the principle of current division, and the circuit in the figure 2.53 is known as the current divider circuit.

Keep in mind:

If N resistors with resistances R_1, R_2, R_3, ..., R_N are connected in parallel across a current source I_s, then the current through the Nth resistor (R_N) will be:

$$I_N = \frac{\frac{1}{R_N}}{\frac{1}{R_1} + \frac{1}{R_2} + ... + \frac{1}{R_N}} I_s$$

SOURCE TRANSFORMATION:

An ideal voltage source has zero internal resistance. In a non-ideal or practical voltage source, internal resistance is present in series with the source, as shown in figure 2.54. The Voltage across the terminals fall as the current through it increases, as shown in figure. The terminal voltage depends on the source current, given as:

$$V_t = V_s - I_t R_s \quad \text{... (i)}$$

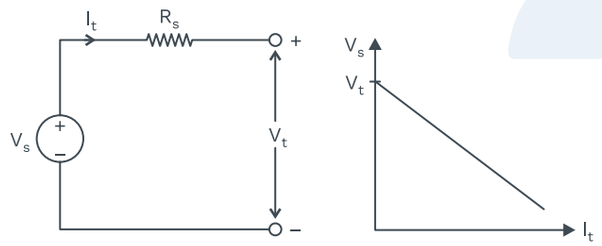


Fig. 2.54 A Practical Voltage Source

An ideal current source has infinite internal resistance. In a practical current source, the internal resistance is parallel with the source as shown in fig. 2.54. Here the magnitude of current falls as the voltage across its terminal increases. The V-I characteristics are shown in the figure below.

Grey Matter Alert!

Conductance in parallel behave as a single conductance whose value is equal to the sum of the individual conductances.

Previous Years' Question

In the circuit shown below, the voltage and current sources are ideal, the voltage (V_{out}) across the current source, in volts, is:

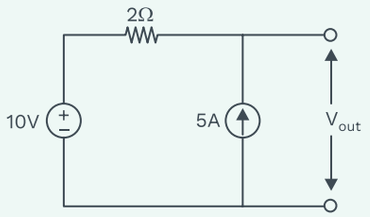


Fig. 2.55

a) 0 **b)** 5
c) 10 **d)** 20
Sol: d)

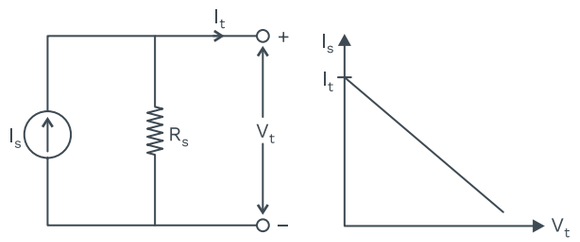


Fig. 2.56 A practical Current Source

The terminal current is given by

$$I_t = I_s - \frac{V_t}{R_s} \qquad \text{... (ii)}$$

Referring to the equations i) and ii), we can say that

$$V_t = I_s R_s - I_t R_s \qquad \text{... (iii)}$$

In order that the figure (2.56) is equivalent to the fig. (2.54), $V_s = I_s R_s$

Hence, if it is required to convert a voltage source V_s in series with an internal resistance R_s into an equivalent current source, it is done by replacing the voltage source with a current source of value $\left(\frac{V_s}{R_s}\right)$ in parallel with a resistance R_s as in figure (2.54). If a current source I_s in parallel with a resistance R_s is to be converted into a voltage source, then it is done by substituting a voltage source ($I_s R_s$) in series with a resistance R_s as in figure 2.57.

Grey Matter Alert!

Transformation of a resistive voltage source to a resistive current source and vice-versa.

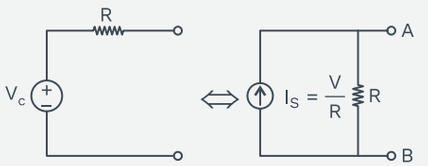


Fig. 2.58

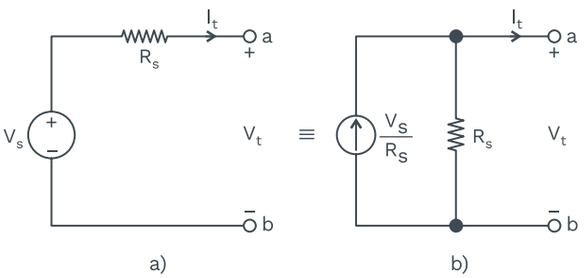


Fig. 2.57 a) Voltage Source
b) Transformed Current Source

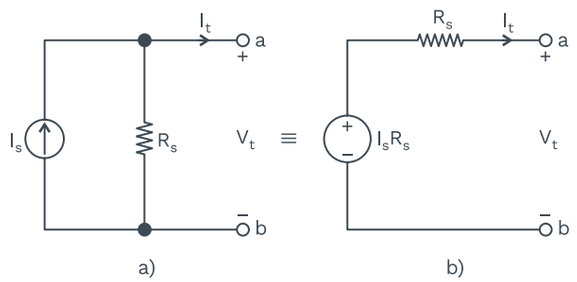


Fig. 2.59 a) Current Source
b) Transformed Voltage Source

Keep in mind:
Source transformation is not valid for ideal voltage and current sources. Source transformation is applicable for the dependent sources also.

SOURCES IN SERIES OR PARALLEL:

In this section, we will study the interconnection of independent voltage and current sources. Four types of interconnection are possible.

Series connection of voltage sources:
Voltage sources connected in series are equivalent to a single voltage source. The voltage of the equivalent voltage source is equal to the sum (according to reference polarity) of voltages of individual sources.
Figures 2.60 a) and 2.60 b) illustrate the series connection of two voltage sources connected with two different reference polarities.

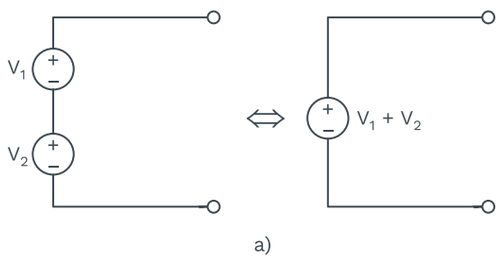


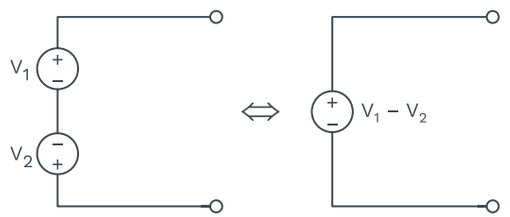
Fig. 2.60

Grey Matter Alert!

Source transformation is applicable for the dependent sources also.

Rack Your Brain

Why source transformation is not valid for ideal voltage and current sources?



b)

Fig. 2.61 Series Connection of Voltage Sources
a) With the Same Reference Polarity
b) With Opposite Reference Polarity

Parallel connection of identical voltage sources:

Voltage sources cannot be connected in parallel unless the two sources have identical voltages. Figure 2.62 illustrates the concept.

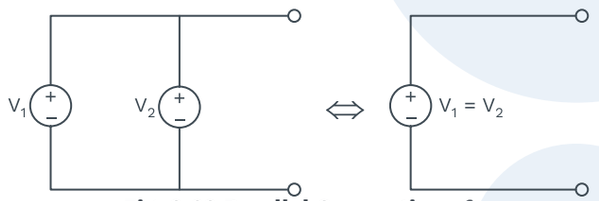
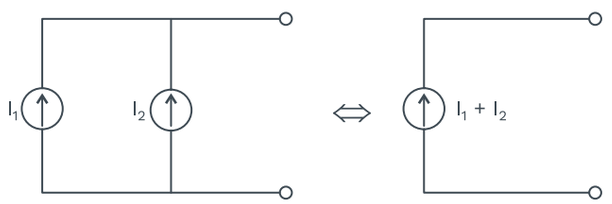


Fig. 2.62 Parallel Connection of Identical Voltage Sources

Parallel connection of current sources:

Current sources connected in parallel are equivalent to a single current source. The current of the equivalent current source is equal to the sum (according to reference direction) of the currents of individual sources. Figure 2.62 a) and 2.62 b) shows the parallel connection of two current sources with the same reference direction and with opposite reference direction.



a)

Fig. 2.63
a) With The Same Reference Direction

Rack Your Brain

How is source transformation useful for circuit analysis?

Grey Matter Alert!

Source transformation cannot be applied in the given circuit.

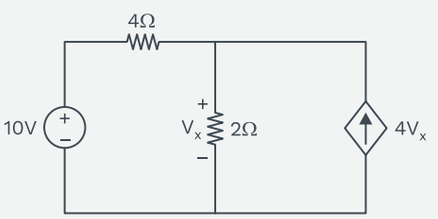
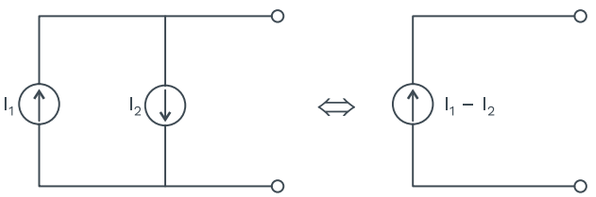


Fig. 2.64



(b)

Fig. 2.65 Parallel Connection of Current Sources b) With Opposite Reference Direction

Series connection of identical current sources:
The current sources cannot be connected in series unless the two sources have identical values. Figure (2.67) illustrates the concept.

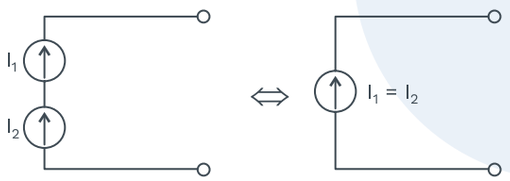


Fig. 2.67 Series Connection of Identical Current Sources

Series-parallel connection of voltage and current sources:
If an ideal voltage source and an ideal current source are connected in series, then the current source output flows through the voltage source without affecting its voltage. Thus, the combination is equivalent to a current source, as shown in figure 2.68.

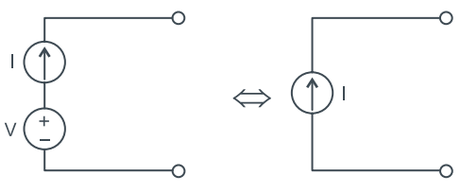


Fig. 2.68 Voltage and Current Sources in Series

Previous Years' Question

In the circuit shown, what value of R will result in I = 4A?

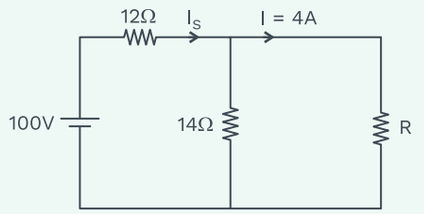


Fig. 2.66

a) 9 Ω **b)** 7 Ω
c) 5.5 Ω **d)** 3.5 Ω
Sol: b)

Grey Matter Alert!

The Resistor in parallel with a voltage source is redundant.

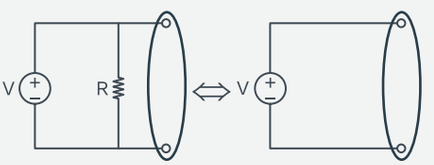


Fig. 2.69

If an ideal voltage source and an ideal current source are connected in parallel, then the voltage across the terminal is the same as the voltage source regardless of the value of current through it. Thus, the combination is equivalent to a voltage source, as shown in figure 2.70.

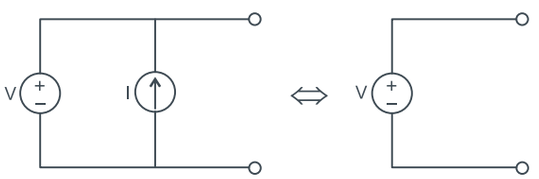


Fig. 2.70 Voltage and Current Sources in Parallel

Example 15:

For the circuit shown in the figure, find the current in 2Ω resistor.

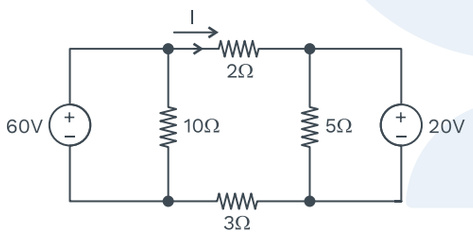
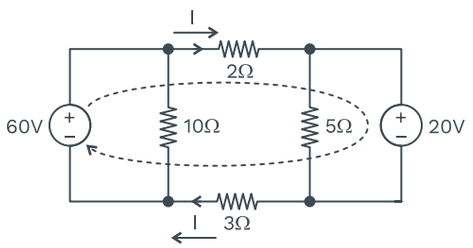


Fig. 2.72

Sol:



Applying KVL in a complete loop:

$60 - 2I - 20 - 3I = 0$

$40 - 5I = 0$

$40 = 5I$

$I = \frac{40}{5}$

$I = 8A$

The current in the 2Ω resistor is 8 amp.

Grey Matter Alert!

The resistor in series with a current source is redundant.

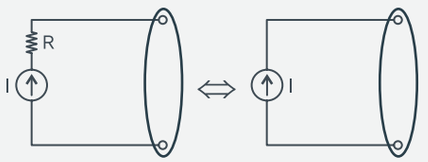


Fig. 2.71

Previous Years' Question

In the portion of a circuit shown, if the heat generated in 5Ω resistance is 10 calories per second, then heat generated by the 4Ω resistance, in calories per second is –

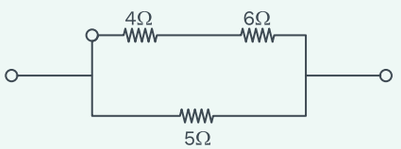


Fig. 2.73

a) 2 **b)** 0.5
c) 3 **d)** 4

Sol: a)

NODAL AND MESH ANALYSIS:

- Nodal and Mesh analysis play an important role in network theory.
- In nodal analysis, first, we have to identify all the different nodes in the network. Identify those nodes where voltages are different.
- Assign arbitrary currents to the branches in which no current is designated. Now apply KCL equations for the branch currents.
- Mesh analysis provides another general procedure for analyzing circuits, using mesh currents as the circuit variables.
- Choosing mesh currents as the circuit variables instead of element currents are more convenient and reduce the number of equations that must be solved simultaneously.

Nodal analysis:

Nodal analysis is a method of determining the voltage (potential difference) between "nodes" (points where elements or branches connect) in an electric circuit.

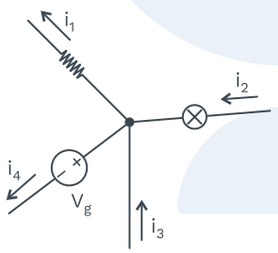


Fig. 2.74 Nodal Analysis

Nodal analysis can be applied for planar as well as non-planar networks.

Key point of nodal analysis:

Nodal analysis can be calculated by KCL + Ohm's law.

Features of nodal analysis:

- Nodal analysis is based on the application of Kirchhoff's current law (KCL).
- For 'n' nodes, there will be 'n – 1' simultaneous equations to solve.
- By solving 'n–1' equations, all the node voltages can be obtained.
- The number of non-reference nodes is equal to the number of nodal equations that can be obtained.

Steps involved in nodal analysis:

The nodal analysis of a circuit involves the following steps:

Step 1:
Select a node as the reference node. Assign voltages V_1, V_2, ..., V_{N-1} to the remaining N – 1 nodes. The voltages are referenced with respect to the reference node.

Step 2:
Apply KCL to each of the N – 1 non-reference nodes. Use Ohm's law to express the branch currents in terms of node voltages.

Step 3:
Solve the resulting simultaneous equations to obtain the unknown node voltages.

Note:
Using node voltage, we can calculate branch voltage in the circuit.

Example 16:
Determine the potential difference between P and Q.

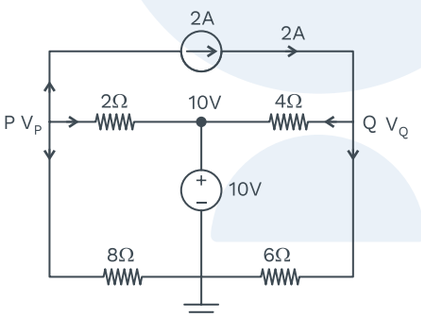


Fig. 2.75

Sol:
Applying KCL at node P-

$$\frac{V_P - 10}{2} + \frac{V_P - 0}{8} + 2 = 0$$

$$4V_P - 40 + V_P + 16 = 0$$

$$5V_P = 24$$

$$V_P = 4.8 \text{ V}$$

Applying KCL at node Q-

$$\frac{V_Q - 10}{4} + \frac{V_Q - 0}{6} - 2 = 0$$

$$3V_Q - 30 + 2V_Q = 24$$

$$5V_Q = 54$$

$V_Q = 10.8$ V

Thus, potential difference

$V_P - V_Q = -6$V

Super node analysis:

Whenever a voltage source (dependent or independent) is connected between the two non-reference nodes, then these two nodes form a generalized node called the super node.

So, the super node can be regarded as a surface enclosing the voltage source and its two nodes.

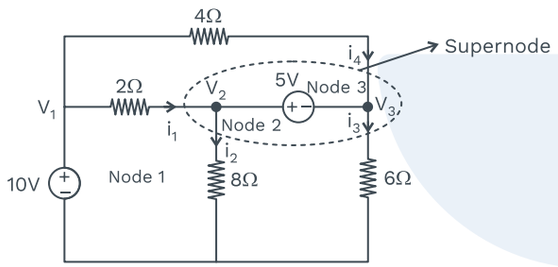


Fig. 2.76 Super node Analysis

In the figure 2.76, 5V source is connected between two non-reference nodes node-2 and node-3.

Thus, node-2 and node-3 form the super node.

Properties of super node:

- The difference between the voltages of two non-reference nodes is known as super node voltage.
- A super node has no voltage of its own.
- A super node requires an application of both KCL and KVL to solve it.
- Any element can be connected in parallel with the voltage source forming the super node.
- A super node satisfies the KCL just like a simple node.

Grey Matter Alert!

At nodes, the current changes and in branches, the current remains same.

Rack Your Brain

Why node analysis is more important as compared to mesh analysis?

Example 17:
Find the node voltage (V_P) at node P.

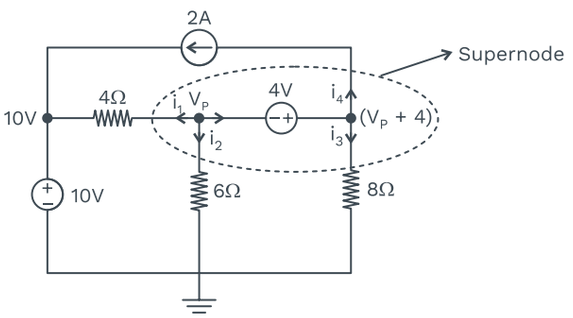


Fig. 2.77 Concept of Super Node

Grey Matter Alert!

The number of non-reference nodes are equal to the number of independent equations that we will derive.

Sol:
Applying KCL-

$$i_1 + i_2 + i_3 + i_4 = 0$$

$$\frac{V_P - 10}{4} + \frac{V_P - 0}{6} + \frac{V_P + 4 - 0}{8} + 2 = 0$$

$$\frac{V_P}{4} - \frac{10}{4} + \frac{V_P}{6} + \frac{V_P}{8} + \frac{1}{2} + 2 = 0$$

$$6V_P - 60 + 4V_P + 3V_P + 12 + 48 = 0$$

$$13V_P = 0$$

$$\therefore \quad V_P = 0 \text{ V}$$

Note:
Never apply KCL at the node where the current source is connected.

Keep in mind:
In the above example, if we want to avoid supernode, then we have to assume a variable current I) through the voltage source, and then we can apply KCL in the circuit.

MESH ANALYSIS:

- Mesh analysis is basically the sum of two laws.
- Mesh analysis = KVL + Ohm's law
- Mesh analysis is used in the circuits to obtain KVL equations, which will help to find the voltage and current in the circuit. It is also known as mesh current method.
- Mesh is a closed-loop which does not consist of any other loop.

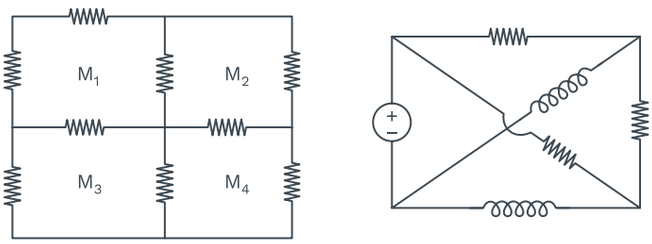


Figure a) Figure b)

Fig. 2.78 Planar Networks

Definitions

Super node: A super node is formed by enclosing a (dependent or independent) voltage source connected between two non-reference nodes and any element connected in parallel with it.

Here, figure 2.78 a) and figure 2.78 b) are planar networks.

When a network is drawn on a plane without any crossover, then the network is called a planar network.

Figure 2.78 b) is planar because it can be represented as-

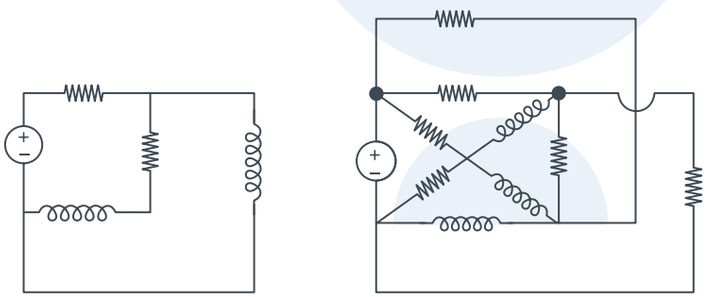


Figure (a) Figure (b)

Fig. 2.79 Non-Planar Networks

The above figure 2.79 b) is non-planar.

Procedure of mesh analysis:

- Identify the total number of meshes in the given network.
- Assign the current direction for each mesh.
- Find the voltage across each branch.
- Develop KVL equation for each mesh.
- By solving KVL equations, find the currents (loop currents).

Grey Matter Alert!

A super node may be regarded as a closed surface enclosing the voltage source and its two nodes.

Example 18:

In the given circuit, the currents i_1, i_2 and i_3 are –

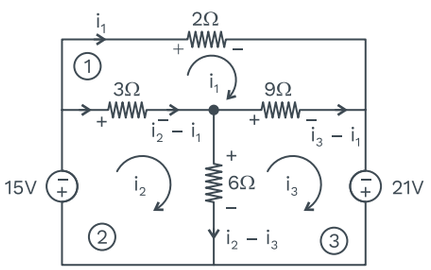


Fig. 2.80

Sol:

Applying KVL in loop 1)–

$9(i_3 - i_1) + 3(i_2 - i_1) - 2i_1 = 0$

$9i_3 - 9i_1 + 3i_2 - 3i_1 - 2i_1 = 0$

$14i_1 - 3i_2 - 9i_3 = 0$ ... (i)

Applying KVL in loop 2)–

$-15 - 3(i_2 - i_1) - 6(i_2 - i_3) = 0$

$3i_1 - 9i_2 + 6i_3 = 15$

$i_1 - 3i_2 + 2i_3 = 5$ ... (ii)

Applying KVL in loop 3)–

$6(i_2 - i_3) - 9(i_3 - i_1) + 21 = 0$

$6i_2 - 6i_3 - 9i_3 + 9i_1 + 21 = 0$

$3i_1 + 2i_2 - 5i_3 = -7$ ... (iii)

On solving these three equations, we get

$i_1 = 3A$

$i_2 = 2A$

$i_3 = 4A$

Key note:

If a circuit has 'n' nodes, 'b' branches, and 'l' independent loops or meshes then,

$$l = b - n + 1$$

Hence, 'l' is the number of an independent simultaneous equations that are required to solve the circuit using mesh analysis.

Previous Years' Question

In the circuit shown below, the node voltage V_A is:

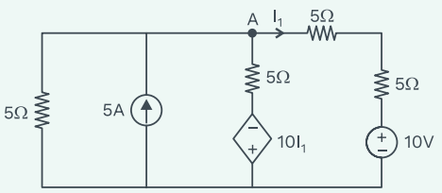


Fig. 2.81

a) 3V **b)** 11V
c) 11.42V **d)** 13V
Sol: c)

Rack Your Brain

When is the super node not applied?

Grey Matter Alert!

Node analysis is applicable for planar and non-planar networks, but mesh analysis is applicable only for planar networks.

SUPERMESH

In nodal analysis, we consider a combination of two nodes as a single node when a voltage source is connected between them. Similarly, we consider the combination of two meshes as a single mesh when a current source is connected between them.

Properties of supermesh:

- A supermesh does not have a current of its own.
- A supermesh requires both the KVL and KCL along with Ohm's law for analysis.

Fun facts:

Super mesh analysis = Ohm's law + KVL + KCL
Mesh analysis = Ohm's law + KVL

Example 19:

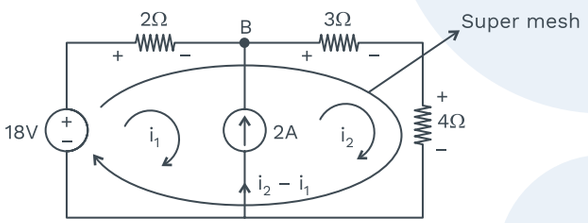


Fig. 2.82

Sol:

Applying KCL at node B:

$$i_2 - i_1 = 2 \quad \ldots (1A)$$

$$\therefore \; i_2 = (i_1 + 2) \quad \ldots (1B)$$

Applying KVL in overall loop:

$$+ 18 - 2i_1 - 3i_2 - 4i_2 = 0$$

$$18 - 2i_1 - 7i_2 = 0 \quad \ldots (2)$$

$\therefore$ Putting value of i_2 from equation 1B) in equation 2) we get:

$$18 - 2i_1 - 7(i_1 + 2) = 0$$
$$18 - 2i_1 - 7i_1 - 14 = 0$$
$$+ 9i_1 = 4$$
$$i_i = \frac{4}{9} A$$

Rack Your Brain

When does super mesh apply?

Definitions

Super mesh: A super mesh results when two meshes have a current source (dependent or independent) in common.

Previous Years' Question

The current in the 1Ω resistor in the network shown below is:

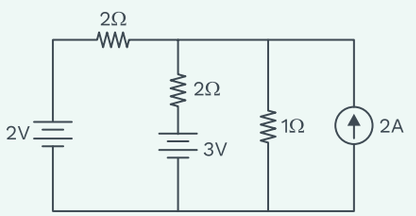


Fig. 2.83

a) 2.00 A **b)** 2.25 A
c) 2.50 A **d)** 2.75 A
Sol: b)

$i_1 = 0.444$ A
and $i_2 = 2.44$ A

Key note:
We cannot apply KVL in a loop when there is a current source present in the loop because the voltage of a current source is a variable and thus it is required to use an extra variable in the mesh. So if we want to apply KVL in each mesh, we have to assume an extra variable as voltage across the current source.

Grey Matter Alert!

The direction of the mesh current is arbitrary – (clockwise or counter-clockwise) – and it does not affect the validity of the solution.

Comparison of nodal and mesh analysis:

Nodal analysis	Mesh analysis
Networks that consist of parallel connected elements like as current source or super nodes are more suitable for nodal analysis.	Networks that consist series-connected elements like voltage sources or super meshes are more suitable for mesh analysis.
A circuit with fewer nodes than meshes are better analyzed by using nodal analysis.	While a circuit with fewer meshes than nodes is better analyzed by mesh analysis.
It is better to prefer a method that results in a smaller number of equation.	If there is a requirement for a branch or mesh currents, it is better to prefer mesh analysis.
Nodal analysis is applicable both for planar and non planar circuit and non- planar circuits.	Mesh analysis is applicable for only planar circuits.
KCL is used in this analysis.	KVL is used in this analysis

Table 1.2

Chapter Summary



- A resistor is a passive element in which the voltage 'v' across it is directly proportional to the current 'i' through it, i.e. a resistor is an element that obeys Ohm's law,

$$V = IR$$

- **Short-circuit:** A short-circuit path has zero resistance (R=0), i.e. maximum current.
- **Open-circuit:** A open-circuit path has infinite resistance (R = ∞)), i.e. zero current.
- Reciprocal of resistance is called conductance, i.e. $G = \frac{1}{R}$
- Two elements are in series when they are connected sequentially, end to end. When elements are in series and the same current flows through them ($i_1 = i_2$). They are in parallel, if they are connected to the same two nodes. Elements in parallel always have the same voltage across them ($v_1 = v_2$).
- If R_1 and R_2 are in series, then its equivalent is given by:

$$R_{eq} = R_1 + R_2$$

- If R_1 and R_2 are in parallel, then its equivalent is given by:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

- **Voltage division rule:** It is applicable in a series network.
  If a circuit has two resistors in series, then voltage across R_1 and R_2 are respectively:

$$V_1 = \frac{R_1}{R_1 + R_2} V, \quad V_2 = \frac{R_2}{R_1 + R_2} V$$

- **Current division rule:** It is applicable in a parallel network.
  If a circuit has two resistors in a parallel, then the current through R_1 and R_2 are, respectively:

$$I_1 = \frac{R_2}{R_1 + R_2} I, \quad I_2 = \frac{R_1}{R_1 + R_2} I$$

- The current in a capacitor is proportional to the rate of change in voltage with respect to time. It is mathematically given by:

$$i = C \frac{dV}{dt}.$$

- When voltage is a constant value (DC value), then the current through a capacitor is zero.
- The voltage across a capacitor is given by:

$$V = \frac{1}{C} \int_{-\infty}^{t} i\,dt = \frac{1}{C} \int_{t_0}^{t} i\,dt + v(t_0)$$

  The voltage across a capacitor cannot change immediately.

- Voltage across an inductor is proportional to the rate of change in current with respect to time.

$$V = L\frac{di}{dt}$$

When current is a constant value (DC value), then the current-voltage across an inductor is zero, i.e. short-circuit.

- The current through an inductor is proportional to the time integral of the voltage across it.

$$i = \frac{1}{L}\int_{-\infty}^{t} v\,dt = \frac{1}{L}\int_{t_0}^{t} v\,dt + i(t_0)$$

The current through an inductor cannot change instantly.

- The energy in capacitors = $\frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C}$
- Energy in inductor = $\frac{1}{2}Li^2$
- Nodal analysis is applicable to both planar and non-planar circuits.
- In the nodal analysis, we have to use KCL for analysis.
- A super node consist, of two non-reference nodes connected by a (dependent or independent) voltage source.
- Mesh analysis is the application applicable only to planar circuits.
- A super-mesh consists of two meshes that have a (dependent or independent) current source in common for both mesh.
- A branch of a network is defined as a group of elements connected in series or parallel combinations and which has two terminals from connections.
- A node is a common point in a network at which two or more branches joint.
- A loop is a set of branches which form a closed path in a network.
  If a network has b number of branches, n numbers of nodes and l number of independent loops, then

$$b = l + n - 1$$

- Kirchhoff's current law (KCL) states that the algebraic sum of all currents entering a leaving a node must be zero. In other words, the sums of the currents directed at any node in a circuit are equal to the sum of currents leaving at the same node.
- Kirchhoff's voltage law (KVL) states that the algebraic sum of voltages around a closed loop in a circuit must be equal to zero. In other statements, the sum of voltage rises is equal to the sum of voltage drops around a closed loop in a circuit.

PRACTICE QUESTIONS

Q1 To move charge from point a to b requires –30 J. Find the voltage V_{ab} if q = 6C.

a) –5V b) 10V c) 3V d) 5V

Q2 In the following network, the value of current I_1, I_2 and I_3, respectively.

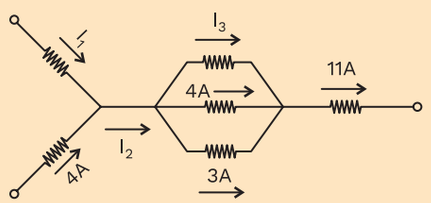


Fig. 2.84

a) 7A, 11A, 0A b) 7A, 11A, 4A c) 11A, 7A, 0A d) 7A, 7A, 4A

Q3 What will be power absorbed or supplied by the given circuit shown below:

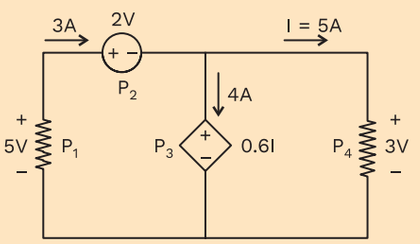


Fig. 2.85

a) $P_1 = 45W, P_2 = 18W$
$P_3 = 120W, P_4 = 150W$

b) $P_1 = 45W, P_2 = -18W$
$P_3 = 12W, P_4 = 15W$

c) $P_1 = -45W, P_2 = 18W$
$P_3 = 12W, P_4 = 15W$

d) $P_1 = 45W, P_2 = 18W$
$P_3 = 12W, P_4 = 15W$

Q4 **Calculate the equivalent resistance R_{ab} in the circuit in the figure.**

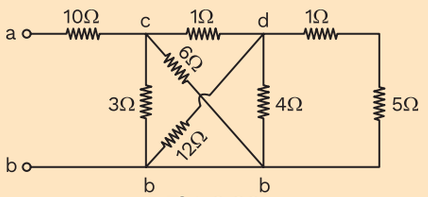


Fig. 2.86

a) 11.2 Ω **b) 11 Ω** **c) 12 Ω** **d) 15 Ω**

Q5 **Calculate G_{eq} in the circuit of the figure.**

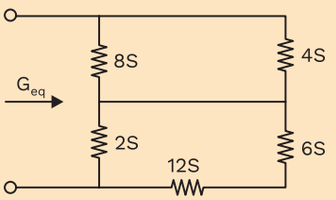


Fig. 2.87

a) 0S **b) 5S** **c) 4.4S** **d) 4S**

Q6 **For the bridge network in the figure, find R_{ab} and i.**

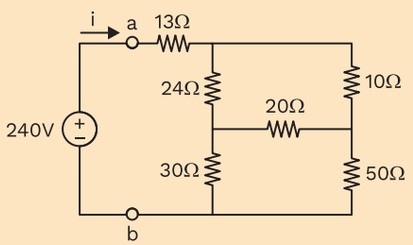


Fig. 2.88

a) 40Ω, 6A **b) 40Ω, 3A** **c) 14Ω, 6A** **d) 40Ω, 9A**

Q7 **For the circuit shown below, the number of branches and nodes are respectively:**

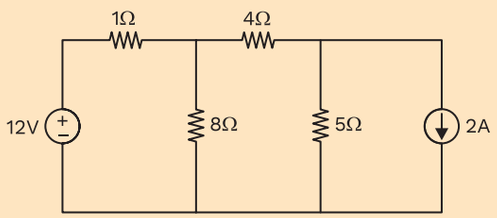


Fig. 2.89

a) 5 branch, 4 node **b) 5 branch, 5 node**
c) 6 branch, 5 node **d) 6 branch, 4 node**

Q8 **Obtain V_1 through V_3 in the circuit of figures.**

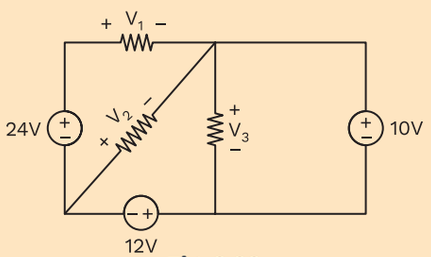


Fig. 2.90

a) 14V, –2V, 10V **b) 14V, –22V, 10V**
c) 2V, –22V, 10V **d) 30V, –22V, 10V**

Q9 **For the circuit shown in the figure, determine all four nodal voltages.**

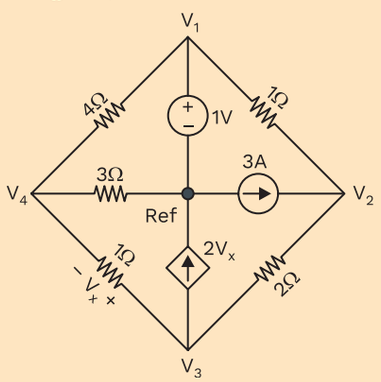


Fig. 2.91

a) $V_1 = 1V, V_2 = 1.25V$
$V_3 = 3.08V, V_4 = 0.951V$

b) $V_1 = 1V, V_2 = 3.08V$
$V_3 = 1.25V, V_4 = 0.951V$

c) $V_1 = 1V, V_2 = 2.25V$
$V_3 = 4.8V, V_4 = 1V$

d) $V_1 = 1V, V_2 = 3.08V$
$V_3 = 1.25V, V_4 = 1V$

Q10 **For the circuit shown in the figure, the equivalent capacitance seen at the terminal a – b is equal to .....μF.**

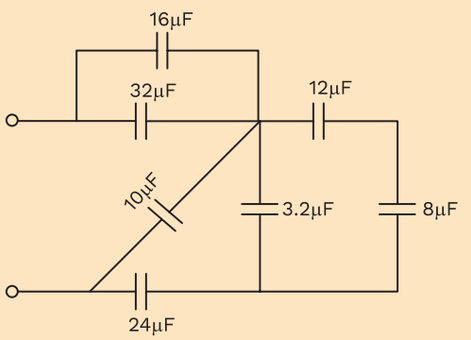


Fig. 2.92

a) 24 b) 12 c) 9 d) 3.2

Q11 **In the figure R_a, R_b and R_c are 20Ω, 10Ω, and 30Ω respectively. The resistance R_1, R_2, R_3 in Ω of an equivalent star connection are**

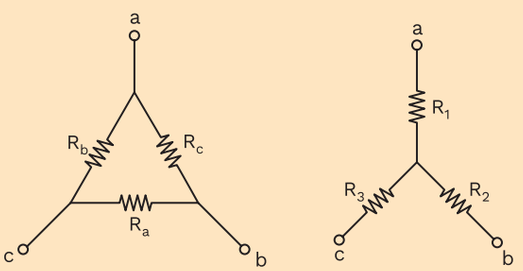


Fig. 2.93

a) 5, 10, $\frac{10}{3}$ b) 5, 5, 10 c) 10, 5, $\frac{10}{3}$ d) 10, 5, 10

Q12 **Determine L_{eq} that may be used to represent the inductive network of figure at the terminals.**

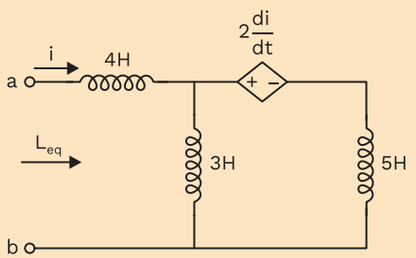


Fig. 2.94

a) 6.1 H **b) 5 H** **c) 6.625 H** **d) 2 H**

Q13 **In the circuit of figure sketch V_0.**

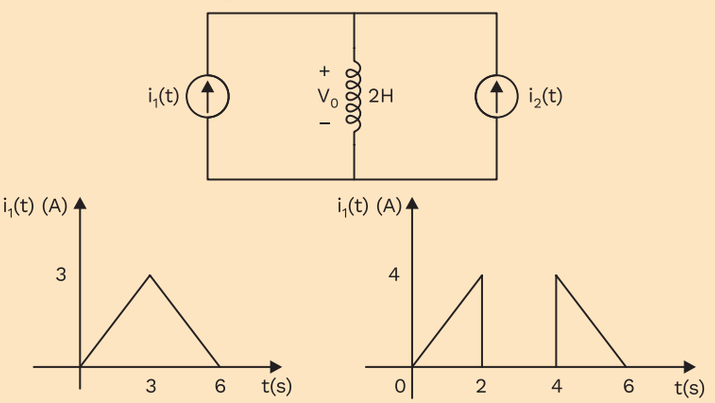
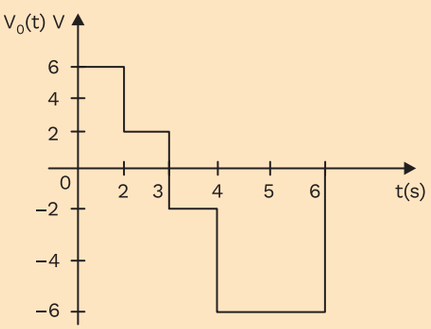
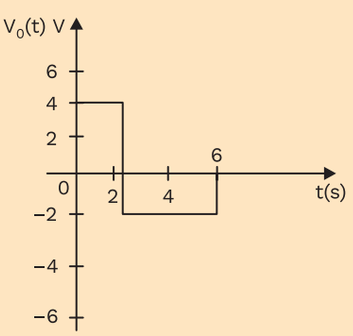


Fig. 2.95

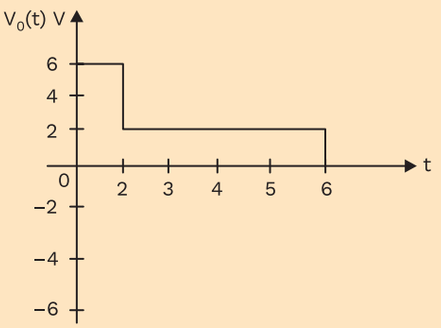
a)



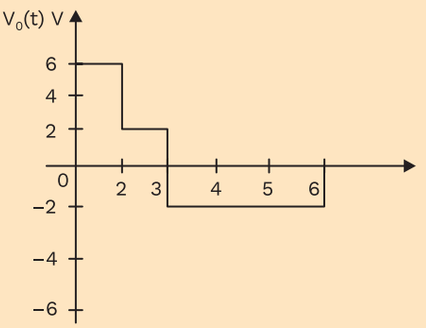
b)



c)



d)



Q14 **For the bridge network in the figure, find i_0 using mesh analysis.**

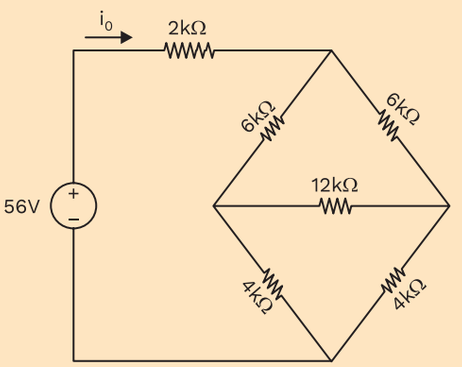


Fig. 2.96

a) 5mA **b) 6A** **c) 8mA** **d) 9A**

Q15 **For the circuit given below, the value of I_b is-**

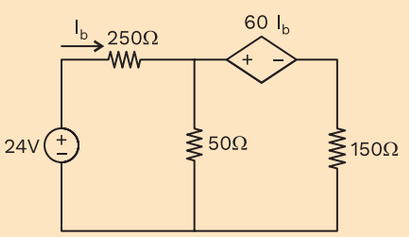


Fig. 2.97

a) 79.34 mA **b) 80 mA** **c) 13 mA** **d) 10 mA**

Answer key:

1)	**a)**	**2)**	**b)**	**3)**	**c)**	**4)**	**a)**	**5)**	**d)**
6)	**a)**	**7)**	**d)**	**8)**	**c)**	**9)**	**b)**	**10)**	**b)**
11)	**a)**	**12)**	**c)**	**13)**	**a)**	**14)**	**c)**	**15)**	**a)**